Term Paper: Neutrinos

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Chapter 1

Introduction and Early Phenomenology

The phenomenon of neutrinos was first proposed by Wolfgang Pauli in 1930 to explain the apparent violation of momentum conservation. For a two body decay, the energy of the daughter particles is fixed. But Pauli observed that in beta decay, the energy of the emitted electron varied continuously, which suggested that either energy conservation was violated or there was an undetected particle carrying away the missing energy. Pauli proposed the existence of a neutral, massless particle, which was termed as "neutrino" by Enrico Fermi in 1934.

The introduction of neutrinos not only solved the problem of energy conservation in beta decay but also took care of the shape of the energy spectrum of the emitted electrons:

$$\frac{d\Gamma}{dE} \propto p_e E_e (E_0 - E_e)\sqrt{(E_0 - E_e)^2 - m_\nu^2} \tag{1.1}$$

where E_0 is the maximum energy of the emitted electron $(Q - m_e - m_\nu)$, E_e is the energy of the emitted electron, p_e is the momentum of the emitted electron and m_ν is the mass of the neutrino. The beta decay experiments till now have only been able to put an upper limit on the "mass" of the electron neutrino, which is around 0.8 eV, with confidence level of 90%[1].

1.1 Discovery of the three flavours of neutrinos

The first direct observation of neutrinos was by Reines and Cowan in 1956, wherein they directed a flux of ν_e (supposed to have come from a beta decay) into a water target. The reaction $\nu_e p \to n e^+$ produced positrons, which annihilated with electrons in the scintillation counters giving two 0.5 MeV photons. The neutrons were absorbed by $CdCl_2$ dissolved in water, which emitted photons within a few μs. The coincidence of these two kinds of photons confirmed the above reaction, and hence the presence of ν_e. Refer section 6 for more details on the detection of neutrinos.

The discovery of the muon neutrino (ν_μ) was made in 1962 by Lederman, Schwartz and Steinberger. They used a beam of high energy protons from the Alternating Gradient Synchrotron at Brookhaven National Laboratory to produce pions, which then decayed into muons and muon neutrinos. The muon neutrinos were detected by observing their interactions in a detector located downstream of the pion production target. The interaction was $\nu_\mu + N \to \mu + N'$. The experiment showed that the muon neutrinos produced

in pion decay did not interact to produce electrons, confirming that they were distinct from electron neutrinos. This was one of the first indications of lepton flavor conservation (that doesn't hold in general, particularly in the case of neutrino oscillations, which will be discussed later).

The discovery of the tau neutrino (ν_τ) was made in 2000 by the DONuT (Direct Observation of the Nu Tau) experiment at CERN[2]. The experiment used a beam of high energy protons to produce a large number of tau leptons, which then decayed into tau neutrinos. The tau neutrinos were detected by observing their interactions in a detector located downstream of the tau production target. The interaction was $\nu_\tau + N \rightarrow \tau + N'$. The main difficulty that resulted in the late discovery of the tau neutrino was that the tau lepton has a very short lifetime (about 2.9×10^{-13} seconds) and decays into multiple particles, making it difficult to identify the tau neutrino interactions.[1]

1.2 Neutrino Mass measurements through direct detections

The experiments with beta decay give an upper bound of 0.8 eV on the mass of "electron" neutrinos, as we have seen above. The mass of "muon" neutrino can be determined by observing the energy of muon emitted in a pion decay at rest. It is a two-body decay, hence monoenergetic, and the energy is a direct function of the mass of the emitted neutrino. Unfortunately the accuracy in the energy measurement is not high enough to be able to determine the ν_μ mass: the current limit is of the order of 100 keV [3]. The mass of ν_τ also has a measured upper bound of a few MeVs. Two things are worth emphasizing here. Firstly, since we now know that the neutrino flavor eigenstates are very different from their mass eigenstates, it is not correct to use the phrase "mass of ν_e /ν_μ /ν_τ", and the above limits are to be taken in a very rough "pre-neutrino-mixing" spirit. Secondly, now that we know that the three mass eigenstates are very close (difference in mass squares is less than 10^{-2} eV^2), the mass limits of keV and MeV are no longer relevant.

1.3 Parity violation in neutrino interactions

The discovery of parity violation in weak interactions was made in 1956 by Chien-Shiung Wu and her collaborators. They conducted an experiment using cobalt-60, which undergoes beta decay. The experiment involved cooling the cobalt-60 sample to very low temperatures and applying a strong magnetic field to align the spins of the cobalt nuclei. They observed that the emitted electrons were preferentially emitted in a direction opposite to the spin of the cobalt nuclei, indicating that the weak interaction responsible for beta decay violates parity symmetry. This was a groundbreaking discovery that challenged the previously held belief that all fundamental interactions were parity conserving. The discovery of parity violation in weak interactions led to the development of the V-A (vector minus axial vector) theory of weak interactions, which successfully explained the observed phenomena and became a cornerstone of the Standard Model of particle physics.

The experiment of Vladimir Telegdi involved observing the angular distribution of electrons from the decay of polarized muons. This also confirmed parity violation.

[1] Some of the facts used in this term paper are taken from lecture notes by Amol Dighe[3]

Both of the above experiments violated parity maximally, meaning that the weak interaction only couples to left-handed particles and right-handed antiparticles. This subsequently led to the V-A model of weak interactions, which postulated the 4-Fermi interaction involving beta decay to be of the form

$$\frac{G_F}{\sqrt{2}}[\bar{p}\gamma^\mu(C_V - C_A\gamma^5)n][\bar{e}\gamma_\mu(1-\gamma^5)\nu_e]$$

Where G_F is the Fermi coupling constant, C_V and C_A are the vector and axial vector coupling constants respectively (due to proton and neutron being composite particles), and γ^μ and γ^5 are the gamma matrices. We now understand that the above interaction is a low energy effective interaction, and the actual interaction is mediated by the exchange of W bosons, which couple only to left-handed particles and right-handed antiparticles.

1.4 Neutrino interactions in the Standard Model

In the Standard Model of particle physics, neutrinos interact with other particles through the weak force, which is mediated by the exchange of W and Z bosons. Neutrinos can interact with other particles in two ways: charged current interactions and neutral current interactions. In charged current interactions, a neutrino interacts with a charged lepton (such as an electron, muon, or tau) through the exchange of a W boson, resulting in the production of a charged lepton and a neutrino. The interaction lagrangian for charged current interactions can be written as:

$$\mathcal{L}_{CC} = -\frac{g}{2\sqrt{2}}\left(\bar{\ell}\gamma^\mu(1-\gamma^5)\nu_\ell W_\mu^- + h.c.\right) \tag{1.2}$$

where g is the weak coupling constant, ℓ represents the charged lepton, ν_ℓ represents the corresponding neutrino, and W_μ^- is the W boson field, h.c. stands for hermitian conjugate. The $(1-\gamma^5)$ term indicates that the interaction only involves left-handed particles and right-handed antiparticles, which is a manifestation of parity violation in weak interactions as $\nu_L = (1-\gamma^5)/2\nu$ is the left-handed component of the neutrino and $\bar{\nu}_R = \bar{\nu}(1+\gamma^5)/2$ is the right-handed component of the neutrino. This also indicates that the W boson interacts with all neutrino flavors equally

In neutral current interactions, a neutrino interacts with another particle (such as a quark or a charged lepton) through the exchange of a Z boson, without changing the flavor of the neutrino. The interaction lagrangian for neutral current interactions can be written as:

$$\mathcal{L}_{NC} = -\frac{g}{2\cos\theta_W}\left(\bar{f}\gamma^\mu(C_V^f - C_A^f\gamma^5)fZ_\mu + h.c.\right) \tag{1.3}$$

where f represents the fermion (quark or charged lepton), C_V^f and C_A^f are the vector and axial vector coupling constants for the fermion f, θ_W is the weak mixing angle, and Z_μ is the Z boson field. The coupling constants C_V^f and C_A^f depend on the type of fermion and are given by:

$$C_V^f = T_3^f - 2Q_f\sin^2\theta_W \tag{1.4}$$

$$C_A^f = T_3^f \tag{1.5}$$

where T_3^f is the third component of the weak isospin of the fermion f, and Q_f is the electric charge of the fermion f. For left handed neutrinos, $T_3^\nu = 1/2$ and $Q_\nu = 0$, so the

coupling constants for neutrinos are:

$$C_V^\nu = \frac{1}{2} \tag{1.6}$$

$$C_A^\nu = \frac{1}{2} \tag{1.7}$$

for right handed neutrinos, $T_3^\nu = 0$ and $Q_\nu = 0$, so the coupling constants for neutrinos are identically zero. This is a manifestation of the fact that right handed neutrinos do not interact with the weak force in the Standard Model.

1.5 Number of neutrino species

The number of neutrino species can be determined from the measurement of the decay width of the Z boson. The Z boson can decay into a fermion-antifermion pair, and by observing the total decay width of the Z boson and subtracting the "visible" contributions from all known fermions (i.e., quarks and charged leptons), we can infer the number of neutrino species that couple to the Z boson. The total width of the Z boson is 2495 MeV [4], and the visible contributions from quarks and charged leptons add 1996 MeV, which leaves 499 MeV for the invisible decay width. Each neutrino species contributes about 167 MeV to the invisible decay width, so we can infer that there are three neutrino species that couple to the Z boson. This is consistent with the three known flavors of neutrinos: electron neutrino, muon neutrino, and tau neutrino. The only way to have more than three neutrino species is to have additional neutrinos that do not couple to the Z boson, which are called sterile neutrinos, or there could be additional neutrinos that are heavier than half the mass of the Z boson, which would not contribute to the invisible decay width of the Z boson. Refer to the section on sterile neutrinos for more details on sterile neutrinos.

Chapter 2

Atmospheric neutrino oscillations

The first evidence that neutrinos had mass came from the observations of neutrinos coming from atmospheric sources. When cosmic rays interact with the Earth's atmosphere, they produce a shower of particles, including pions and kaons, which then decay into muons and muon neutrinos. The muons further decay into electrons and electron neutrinos. Therefore, we expect a certain ratio of muon neutrinos to electron neutrinos in the atmospheric neutrino flux. However, experiments such as Super-Kamiokande observed a deficit of muon neutrinos compared to electron neutrinos, which could be explained by the phenomenon of neutrino oscillations where neutrinos can change their flavor as they propagate through space. This implies that neutrinos have mass and that the flavor eigenstates are not the same as the mass eigenstates.

2.1 Observations and phenomenology

When cosmic rays interact with the Earth's atmosphere, they produce a pions and kaons, and these particles undergo the following decay chains:

$$\pi^{\pm} \to \mu^{\pm} + \nu_{\mu}(\bar{\nu}_{\mu}), K^{\pm} \to \mu^{\pm} + \nu_{\mu}(\bar{\nu}_{\mu}) \tag{2.1}$$

$$\mu^{\pm} \to e^{\pm} + \nu_{e}(\bar{\nu}_{e}) + \bar{\nu}_{\mu}(\nu_{\mu}) \tag{2.2}$$

Therefore, as long as the detectors don't differentiate between the neutrino and anti-neutrino, we expect a certain ratio of muon neutrinos to electron neutrinos in the atmospheric neutrino flux, which is approximately 2:1. However this is not what experiments observed. They observed the muon neutrino flux to be significantly lower than the expected value, while the electron neutrino flux was consistent with expectations. This meant that something was missing in the model. However, the deficit of muon neutrinos could be explained by the model dependence of the expected flux, which is based on the hadronic interaction models and the cosmic ray flux models. Therefore, it was not clear whether the deficit of muon neutrinos was due to new physics or due to uncertainties in the models.

The crucial evidence for oscillations of neutrinos came from the observation of the zenith angle dependence of the muon neutrino flux. The Super-Kamiokande experiment (which was a water based Cherenkov detector) observed that the deficit of muon neutrinos was more pronounced for neutrinos coming from below (i.e., neutrinos that had traveled a longer distance through the Earth) compared to neutrinos coming from above (i.e., neutrinos that had traveled a shorter distance through the atmosphere)[5]. The main feautures of the observations were:

- Under the assumption of no oscillations, the flux of neutrinos v.s. the $\cos\theta$ (where θ is the zenith angle) for sub-GeV neutrinos should be flat,i.e., the flux for both the neutrino flavours is isotropic. For Multi-GeV neutrinos, the flux should be higher at $\cos\theta = 0$ (i.e., neutrinos coming from the horizon), as the cosmic rays and muons and kaons have to travel a longer distance through the atmosphere, which increases the probability of decay and hence the production of neutrinos. This effect is absent for sub-GeV neutrinos, as the parent particles have a very short lifetime in Earth's frame and hence decay quickly, making the distribution isotropic.

- The expected number of events (under the assumption of no oscillations) is symmetric in $\cos\theta$ for both the neutrino flavours, module the effects of Earth's magnetic field, which is well understood.

- Observations match the expectations for electron neutrinos, at all energy scales.

- The flux of muon neutrinos is always lower than the expected value for sub-GeV neutrinos.The deficit of muon neutrinos is more pronounced for neutrinos coming from below (i.e., neutrinos that had traveled a longer distance through the Earth, i.e., $\cos\theta < 0$) compared to neutrinos coming from above (i.e., neutrinos that had traveled a shorter distance through the atmosphere).

- For Multi-GeV neutrinos, the flux of muon neutrinos is consistent with expectations for neutrinos coming from above, while the deficit of muon neutrinos is more pronounced for neutrinos coming from below.

It can be immediately be concluded from these observations that whatever process is responsible for the deficit of muon neutrinos, it must be a function of the distance traveled by the neutrinos as well as the energy of the neutrinos. Moreover, the process must not affect the electron neutrinos (atleast to a significant extent). The only known process that can explain these observations is the phenomenon of neutrino oscillations, which is a quantum mechanical phenomenon where neutrinos can change their flavor as they propagate through space. The phenomenon of neutrino oscillations was first proposed by Bruno Pontecorvo in 1957, and it was further developed by Ziro Maki, Masami Nakagawa and Shoichi Sakata in 1962, long before the experimental evidence for neutrino oscillations was observed. The phenomenon of neutrino oscillations implies that neutrinos have mass and that the flavor eigenstates are not the same as the mass eigenstates.

2.2 Two neutrino oscillation in vacuum

The phenomenon of neutrino oscillations can be understood by considering a simple two-flavor model, where we only consider two flavors of neutrinos, say ν_μ and ν_τ. In this model, the flavor eigenstates are related to the mass eigenstates by a unitary transformation:

$$\begin{pmatrix} \nu_\mu \\ \nu_\tau \end{pmatrix} = \begin{pmatrix} \cos\theta & \sin\theta \\ -\sin\theta & \cos\theta \end{pmatrix} \begin{pmatrix} \nu_1 \\ \nu_2 \end{pmatrix} \tag{2.3}$$

where ν_1 and ν_2 are the mass eigenstates with masses m_1 and m_2 respectively, and θ is the mixing angle. In general, the mixing can be given as:

$$|\nu_\alpha\rangle = \sum_i U_{\alpha i} |\nu_i\rangle \tag{2.4}$$

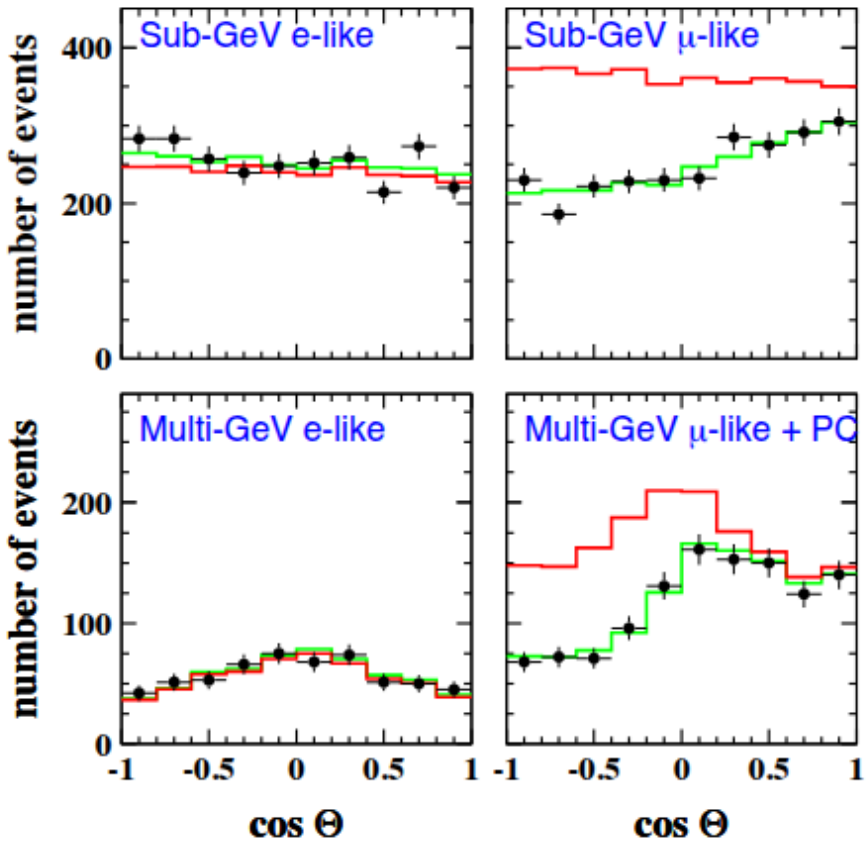

Figure 2.1: The flux of atmospheric neutrinos as a function of the zenith angle. The red line denotes the expected flux under the assumption of no oscillations, while the green line with black dots denote the observed flux. The top panel is for sub-GeV neutrinos, while the bottom panel is for Multi-GeV neutrinos. The left panel is for electron neutrinos, while the right panel is for muon neutrinos. [3]

where $|\nu_\alpha\rangle$ is the flavor eigenstate, $|\nu_i\rangle$ is the mass eigenstate, and $U_{\alpha i}$ is the element of the unitary mixing matrix. When neutrinos are produced in a weak interaction, they are produced in a flavor eigenstate, which is a superposition of the mass eigenstates. As the neutrinos propagate through space, the mass eigenstates evolve differently due to their different masses, leading to a change in the flavor composition of the neutrino beam.

The energy eigenvalues of the system while propagating is given by:

$$E_i = \sqrt{p^2 + m_i^2} \approx p + \frac{m_i^2}{2p}$$

where p is the momentum of the neutrino, and we have used the approximation that the neutrino is ultra-relativistic (i.e., $p \gg m_i$). Interference can happen between two mass eigenstates only if they travel coherently, which means that the momenta of the two mass eigenstates should be equal, i.e., $p_1 = p_2 = p$. Now, we know that the eigenstate $|\nu_i\rangle$ evolves in time as $|\nu_i(t)\rangle = e^{-iE_i t}|\nu_i(0)\rangle$. Since the momenta of the two mass eigenstates are equal, we can ignore the common phase factor e^{-ipt}, and hence the effective energy eigenvalues of the system can be written as:

$$E_i = \frac{m_i^2}{2p}$$

If a neutrino is produced in a flavor eigenstate $|\nu_\alpha\rangle$ at time $t = 0$, then it can be expressed as a superposition of the mass eigenstates as follows:

$$|\nu_\alpha\rangle = \cos\theta|\nu_1\rangle + \sin\theta|\nu_2\rangle \tag{2.5}$$

As the neutrino propagates through space, the mass eigenstates evolve differently due to their different masses, leading to a change in the flavor composition of the neutrino beam. After a time t, the state of the neutrino can be written as:

$$|\nu(t)\rangle = \cos\theta e^{-iE_1 t}|\nu_1\rangle + \sin\theta e^{-iE_2 t}|\nu_2\rangle \tag{2.6}$$

The probability of the neutrino being detected as a different flavor eigenstate $|\nu_\beta\rangle$ at time t can be calculated as follows:

$$P_{\alpha\to\beta}(t) = |\langle\nu_\beta|\nu(t)\rangle|^2 \tag{2.7}$$

$\nu_\beta = -\sin\theta|\nu_1\rangle + \cos\theta|\nu_2\rangle$, hence

$$\begin{aligned}
\langle\nu_\beta|\nu(t)\rangle &= -\sin\theta\cos\theta e^{-iE_1 t} + \sin\theta\cos\theta e^{-iE_2 t} \\
&= \sin(2\theta)(e^{-iE_1 t} - e^{-iE_2 t})/2 \\
&= \sin(2\theta)e^{-i(E_1+E_2)t/2}(e^{-i(E_1-E_2)t/2} - e^{i(E_1-E_2)t/2})/2 \\
&= \sin(2\theta)e^{-i(E_1+E_2)t/2}(-2i)\sin((E_1-E_2)t/2)/2 \\
&= \sin(2\theta)\sin((E_1-E_2)t/2)
\end{aligned}$$

This implies that the probability of the neutrino being detected as a flavor eigenstate $|\nu_\beta\rangle$ at time t is given by:

$$P_{\alpha\to\beta}(t) = \sin^2(2\theta)\sin^2\left(\frac{(E_1 - E_2)t}{2}\right) \tag{2.8}$$

Since the neutrino is ultra-relativistic, we can use the effective energy eigenvalues of the system, which are given by $E_i = m_i^2/2p$ and write $t = L$ (the distance traveled) and $p = E$ to rewrite the above equation as follows:

$$P_{\alpha\to\beta}(t) = \sin^2(2\theta)\sin^2\left(\frac{\Delta m^2 L}{4E}\right) \tag{2.9}$$

where $\Delta m^2 = m_2^2 - m_1^2$ is the difference in the squares of the masses of the two neutrino mass eigenstates and E is the dynamic energy of the neutrino. The survival probability of the neutrino being detected as the same flavor eigenstate $|\nu_\alpha\rangle$ at time t can be calculated as follows:

$$P_{\alpha\to\alpha}(t) = 1 - \sin^2(2\theta)\sin^2\left(\frac{\Delta m^2 L}{4E}\right) \tag{2.10}$$

If we express the above equation in experimental units, we get:

$$P_{\alpha\to\beta}(t) = \sin^2(2\theta) sin^2\left(1.27\frac{\Delta m^2(\mathrm{eV}^2)L(\mathrm{km})}{E(\mathrm{GeV})}\right) \tag{2.11}$$

2.3 Explaining the atmospheric neutrino observations with neutrino oscillations

We can see that the wavelength of oscillation, $\lambda \sim \frac{E}{\Delta m^2}$. Hence the low energy muon neutrinos have a shorter wavelength of oscillation compared to the high energy neutrinos, shorter than the length of the atmosphere. Therefore, the low energy neutrinos will have a more pronounced deficit compared to the high energy neutrinos as they oscillate into tau neutrinos, which is consistent with the observations. However, for high energy neutrinos, the wavelength of oscillation is longer than the length of the atmosphere, hence we don't expect a significant deficit for high energy neutrinos coming from above, which is also consistent with the observations. For neutrinos coming from below, the distance traveled by the neutrinos is much longer (of the order of the diameter of the Earth), hence we expect a significant deficit for high energy neutrinos coming from below, which is also consistent with the observations. Therefore, the phenomenon of neutrino oscillations can explain all the observed features of atmospheric neutrino flux, assuming that the electron neutrinos do not oscillate (or at least don't oscillate to a significant extent) and that the muon neutrinos oscillate to tau neutrinos.

The data would be fitted to the above oscillation formula to extract the values of the mixing angle θ and the mass squared difference Δm^2. The best fit values for the mixing angle and the mass squared difference from the Super-Kamiokande experiment are $\sin^2(2\theta) > 0.92$ and $\Delta m^2_{atm} = (1.5 - 3.4) \times 10^{-3}$ eV^2 at 90% confidence level.

Note that the oscillation amplitude is maximal, which means that the mixing angle θ is close to 45°. This is in contrast to the quark sector, where the mixing angles are small.

2.4 Ruling out other hypotheses

While oscillations of muon neutrino into tau neutrino explains the observations, since tau neutrinos are very difficult to observe (it was only in 2017 that the first direct detection of tau neutrinos was made by the ICECUBE experiment), the observations could also be

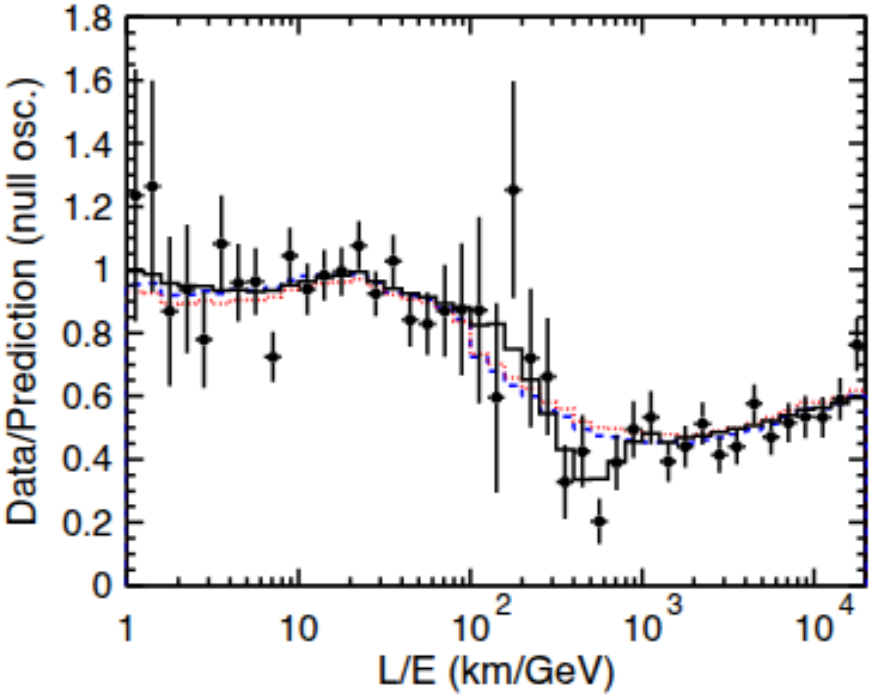

Figure 2.2: The variation of the muon neutrino flux with the L/E ratio as measured by the Super-Kamiokande experiment.

explained by the oscillations of muon neutrino into sterile neutrino, which is a hypothetical neutrino that does not interact with the weak force. Refer the section on sterile neutrinos for more details on sterile neutrinos. The smoking gun evidence for neutrino oscillations would be the observation of the variation of the muon neutrino flux with the L/E ratio, which would show a decrease and then an increase in the survival probability as a function of L/E. The superKamiokande experiment has provided such evidence. They announced a dip in their L/E spectrum (refer to Fig. 2.2) In order to confirm the atmospheric neutrino solution independently, one can produce ν_μ beams with known energies and fluxes and detect the ν_μ component left in them after traveling a certain distance. This experiment will not be affected by any model dependence in the calculation of atmospheric neutrino fluxes. The "short baseline" experiment K2K, where ∼GeV ν_μ were produced and allowed to travel ∼250 km to a detector, has confirmed the solution of the atmospheric neutrino anomaly [6].

Chapter 3

Two neutrino oscillations in matter

Almost always, neutrinos propagate through matter, and the presence of matter can affect the oscillation probabilities. This is known as the Mikheyev-Smirnov-Wolfenstein (MSW) effect and has profound impact on the phenomenology of neutrino oscillations.

3.1 Effective Potentials

When neutrinos propagate through matter, they can interact with the electrons, protons and neutrons in the matter through the weak force. These interactions can be described by an effective potential that modifies the Hamiltonian of the neutrino system. The effective potential for neutral current interactions is the same for all neutrino flavors and is given by:

$$V_{NC} = -\frac{G_F}{\sqrt{2}} N_n \tag{3.1}$$

where N_n is the number density of nucleons in the matter. The effective potential for charged current interactions is only relevant for electron neutrinos (since ordinary matter has electrons and not muons or taus) and is given by:

$$V_{CC} = \sqrt{2} G_F N_e \tag{3.2}$$

where N_e is the number density of electrons in the matter. Note that only the v_e component is affected by the charged current interaction.

3.2 Neutrino oscillation in matter of constant density

The presence of the effective potentials modifies the Hamiltonian of the neutrino system. Consider a two-flavor system of neutrinos, say ν_e and ν_x (where ν_x can be either ν_μ or ν_τ or any linear combination of them). The Hamiltonian of the system in the flavor basis (ν_e, ν_x) can be written as:

$$H_f = U \begin{pmatrix} \frac{m_1^2}{2E} & 0 \\ 0 & \frac{m_2^2}{2E} \end{pmatrix} U^\dagger + \begin{pmatrix} V_{CC} + V_{NC} & 0 \\ 0 & V_{NC} \end{pmatrix} \tag{3.3}$$

where U is the mixing matrix that relates the flavor eigenstates to the mass eigenstates in absence of matter. Substituting the expressions for the unitary mixing matrix in terms

of θ, we can rewrite the Hamiltonian as follows:

$$H_f = \begin{pmatrix} \frac{m_1^2}{2E}\cos^2\theta + \frac{m_2^2}{2E}\sin^2\theta + V_{CC} + V_{NC} & \left(\frac{m_2^2}{2E} - \frac{m_1^2}{2E}\right)\sin\theta\cos\theta \\ \left(\frac{m_2^2}{2E} - \frac{m_1^2}{2E}\right)\sin\theta\cos\theta & \frac{m_1^2}{2E}\sin^2\theta + \frac{m_2^2}{2E}\cos^2\theta + V_{NC} \end{pmatrix} \tag{3.4}$$

Since subtracting same term from the diagonal elements of the Hamiltonian does not change the oscillation probabilities, we can subtract $V_{NC} + (m_1^2 + m_2^2)/4E$ to get the effective Hamiltonian as follows:

$$H_f = \frac{1}{4E}\begin{pmatrix} -\frac{\Delta m^2}{4E}\cos 2\theta + 2A & \frac{\Delta m^2}{4E}\sin 2\theta \\ \frac{\Delta m^2}{4E}\sin 2\theta & \frac{\Delta m^2}{4E}\cos 2\theta \end{pmatrix} \tag{3.5}$$

where $A = 2EV_{CC}$ is the matter potential. The effective mixing angle in matter, θ_m, can be defined as the angle of the 2x2 unitary matrix that diagonalizes the effective Hamiltonian in matter, and the effective mass squared difference in matter, Δm_m^2, can be defined as the difference in the eigenvalues of the effective Hamiltonian in matter. The eigenvalues of the effective Hamiltonian in matter can be calculated as follows:

$$\lambda_\pm = \frac{m_{2m,1m}^2}{2E} = \frac{1}{2E}\left(\frac{A}{2} \pm \frac{1}{2}\sqrt{(\Delta m^2\cos 2\theta - A)^2 + (\Delta m^2 \sin 2\theta)^2}\right) \tag{3.6}$$

where m_{2m} and m_{1m} are the effective masses of the two neutrino mass eigenstates in matter. The effective mass squared difference in matter can be calculated as follows:

$$\Delta m_m^2 = m_{2m}^2 - m_{1m}^2 = \sqrt{(\Delta m^2\cos 2\theta - A)^2 + (\Delta m^2 \sin 2\theta)^2} \tag{3.7}$$

The effective mixing angle in matter would be the angle of the 2x2 unitary matrix that diagonalizes the effective Hamiltonian in matter, which can be calculated as follows:

$$\tan 2\theta_m = \frac{\Delta m^2 \sin 2\theta}{\Delta m^2 \cos 2\theta - A} \tag{3.8}$$

The significance of the matter effect can be understood by looking at the expression for the effective mixing angle in matter. If the matter potential A is equal to $\Delta m^2 \cos 2\theta$, then the effective mixing angle in matter becomes 45°, which means that the mixing is maximal in matter, even if the mixing angle in vacuum is small. This is known as the MSW resonance condition and can lead to a significant enhancement of neutrino oscillations in matter.

As we can see, if the matter density is small, i.e. $A \ll \Delta m^2 \cos 2\theta$, then $\tan(2\theta_m) \approx \tan(2\theta)$, which means that the effective mixing angle in matter is approximately equal to the mixing angle in vacuum, and the oscillation probabilities are approximately the same as in vacuum. However, if the matter density is large, i.e. $A \gg \Delta m^2$, then $\tan(2\theta_m) \approx 0^-$, which means that $2\theta_m \approx 180°$, i.e., mixing is suppressed in matter.

The effective mass eigenvalues in matter as a function of the matter potential A can be visualized in a level crossing diagram. At extremely high densities (i.e., when $A \gg \Delta m^2$), the electron neutrino ν_e corresponds to the heavier mass eigenstate in matter ν_{2m}, while the other flavor neutrino ν_x corresponds to the lighter mass eigenstate in matter ν_{1m}.

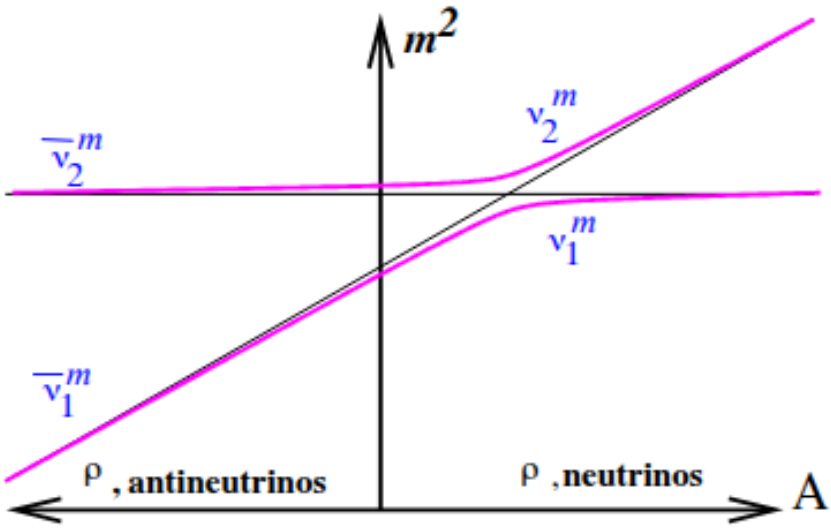

Figure 3.1: The level crossing diagram showing the effective mass eigenvalues in matter as a function of the matter potential A which is a proxy for matter density. The $A < 0$ region corresponds to anti-neutrinos, while the $A > 0$ region corresponds to neutrinos.

3.3 Ruling out the sterile neutrino from atmospheric neutrino oscillations

The key point to note from the above result is that in the presence of matter $\theta \to \theta_m$ and $\Delta m^2 \to \Delta m_m^2$. Hence the oscillation probabilities in matter can be calculated by substituting θ with θ_m and Δm^2 with Δm_m^2 in the oscillation probabilities in vacuum.

The atmospheric neutrino observations can be explained by the oscillations of muon neutrinos into tau neutrinos, which is consistent with the Standard Model of particle physics. However, since tau neutrinos are very difficult to observe, the observations could also be explained by the oscillations of muon neutrinos into sterile neutrinos, which are hypothetical neutrinos that do not interact with the weak force. The presence of matter effects can help us distinguish between these two hypotheses. Since sterile neutrinos do not interact with matter, the oscillations of muon neutrinos into sterile neutrinos would be affected by matter effects (as only the muon neutrinos interact with matter), while the oscillations of muon neutrinos into tau neutrinos would not be affected by matter effects (as both muon neutrinos and tau neutrinos interact with matter in the same way with same effective potential) and the above framework of neutrino oscillations would apply. This is particularly relevant for high energy neutrinos, where $\Delta m^2/(2E) < A$, and matter effects are significant. The observation that no such matter effects are present even in the high energy neutrino events rules out the hypothesis of muon neutrinos oscillating into sterile neutrinos as the explanation of the atmospheric neutrino data.

3.4 Neutrino Oscillations in varying matter density

In many cases, neutrinos propagate through matter with varying density, such as the Sun or the Earth. In such cases, the effective potential A is a function of the position of the neutrino, and the (effective) Hamiltonian of the system becomes time-dependent. The extent of mixing depends not only the matter density but also on the rate of change of the matter density.

The neutrinos produced by the nuclear reactions inside stars start their life in regions of very high densities and travel outwards to the surface of the stars and the interstellar space. As the matter density changes, so do the neutrino mass eigenstates ν_{im}. The propagation of neutrinos can be conveniently analyzed in the basis of these mass eigenstates. In many physical situations, the mass eigenstates may become decoherent, in which case their propagation may be studied independently. If the matter density changes slowly enough, then the neutrino system can be treated as an adiabatic system, and the neutrinos will remain in the same instantaneous mass eigenstate as they propagate through matter. This is known as the adiabatic approximation and can lead to a significant enhancement of neutrino oscillations in matter, even if the mixing angle in vacuum is small.

Let θ_m be the local angle of the 2×2 unitary matrix that diagonalizes the effective Hamiltonian in matter at a given position. We have:

$$\begin{pmatrix} \nu_{1m} \\ \nu_{2m} \end{pmatrix} = U^\dagger(\theta_m) \begin{pmatrix} \nu_e \\ \nu_\mu \end{pmatrix} \tag{3.9}$$

where ν_{1m} and ν_{2m} are the local mass eigenstates in matter. Differentiating the above equation with respect to time, we get:

$$i\frac{d}{dt}\begin{pmatrix} \nu_{1m} \\ \nu_{2m} \end{pmatrix} = i\frac{d}{dt}\left(U^\dagger(\theta_m)\right)\begin{pmatrix} \nu_e \\ \nu_\mu \end{pmatrix} + U^\dagger(\theta_m) i\frac{d}{dt}\begin{pmatrix} \nu_e \\ \nu_\mu \end{pmatrix} \tag{3.10}$$

$$i\frac{d}{dt}\begin{pmatrix} \nu_{1m} \\ \nu_{2m} \end{pmatrix} = i\frac{d}{dt}\left(U^\dagger(\theta_m)\right)U(\theta_m)\begin{pmatrix} \nu_{1m} \\ \nu_{2m} \end{pmatrix} + U^\dagger(\theta_m) H_f U(\theta_m)\begin{pmatrix} \nu_{1m} \\ \nu_{2m} \end{pmatrix} \tag{3.11}$$

Putting $U^\dagger(\theta_m) H_f U(\theta_m) = \mathrm{diag}(m_{1m}^2/2E, m_{2m}^2/2E)$ then, upon calculation of the first term, we get:

$$i\frac{d}{dt}\begin{pmatrix} \nu_{1m} \\ \nu_{2m} \end{pmatrix} = \begin{pmatrix} \frac{m_{1m}^2}{2E} & -i\dot{\theta}_m \\ i\dot{\theta}_m & \frac{m_{2m}^2}{2E} \end{pmatrix}\begin{pmatrix} \nu_{1m} \\ \nu_{2m} \end{pmatrix} \tag{3.12}$$

where $\dot{\theta}_m$ is the time derivative of the local mixing angle in matter (which also is the derivative with respect to the position of the neutrino, as the neutrino is propagating through matter with speed roughly 1). The adiabatic approximation is valid when the off-diagonal terms in the above equation are small compared to the difference in the diagonal terms, i.e., when

$$2\dot{\theta}_m \ll |\Delta m_m^2|/(4E)$$

We have

$$
\begin{aligned}
\tan 2\theta_m &= \frac{\Delta m^2 \sin 2\theta}{\Delta m^2 \cos 2\theta - A} \\
\Rightarrow 2\sec^2 2\theta_m \dot{\theta}_m &= \frac{\Delta m^2 \sin 2\theta}{(\Delta m^2 \cos 2\theta - A)^2} \frac{dA}{dx} \\
&= \tan^2 2\theta_m \frac{1}{\Delta m^2 \sin 2\theta} \frac{dA}{dx} \\
\Rightarrow 2\dot{\theta}_m &= \frac{\tan^2 2\theta_m}{1 + \tan^2 2\theta_m} \frac{1}{\Delta m^2 \sin 2\theta} \frac{dA}{dx} \\
\Rightarrow 2\dot{\theta}_m &= \frac{(\Delta m^2 \sin 2\theta)^2}{(\Delta m^2 \cos 2\theta - A)^2 + (\Delta m^2 \sin 2\theta)^2} \frac{1}{\Delta m^2 \sin 2\theta} \frac{dA}{dx} \\
\Rightarrow 2\dot{\theta}_m &= \frac{\Delta m^2 \sin 2\theta}{(\Delta m_m^2)^2} \frac{dA}{dx}
\end{aligned}
$$

This implies that the adiabatic approximation is valid when

$$
\frac{\Delta m^2 \sin 2\theta}{(\Delta m_m^2)^2} \frac{dA}{dx} \ll \frac{|\Delta m_m^2|}{4E} \tag{3.13}
$$

Clearly, the equality is the weakest when Δm_m^2 is minimum, which happens at the MSW resonance condition, i.e., when $A = \Delta m^2 \cos 2\theta$. At resonance, $\Delta m_m^2 = \Delta m^2 \sin 2\theta$, and hence the adiabatic approximation is valid when

$$
\begin{aligned}
\gamma_{res} =& \frac{(\Delta m_m^2)^3}{2E\Delta m^2 \sin 2\theta}\Big|_{\text{res}} \left|\frac{dA}{dx}\right|_{res}^{-1} \gg 1 \\
=& \frac{(\Delta m^2 \sin 2\theta)^3}{2E\Delta m^2 \sin 2\theta} \frac{A}{\Delta m^2 \cos 2\theta} \left|\frac{dA}{dx}\right|_{res}^{-1} \\
=& \frac{\Delta m^2 \sin^2 2\theta}{2E \cos 2\theta} \left|\frac{1}{A}\frac{dA}{dx}\right|_{res}^{-1} \gg 1
\end{aligned}
$$

This condition is a sufficient condition for the adiabatic approximation to be valid. If the adiabatic approximation is valid, then the neutrinos will almost remain in the same instantaneous mass eigenstate as they propagate through matter with exponentially small probability of jumping to the other mass eigenstate. The jump probability can be calculated using the Landau-Zener formula, which gives the probability of a non-adiabatic transition between two energy levels in a time-dependent Hamiltonian. The Landau-Zener formula for the jump probability is given by:

$$
P_{jump} = e^{-\pi\gamma_{res}/2} \tag{3.14}
$$

This formula gives us the probability of transition when A is varied linearly with respect to the position of the neutrino passing through the resonance point. Note that it depends on Δm^2, mixing angle, the variation of density profile encountered by the neutrinos at resonance, as well as the neutrino energy. Strictly speaking, the above formula is only valid for a linear change in the matter potential A with respect to the position of the neutrino. However, it can be shown that the above formula is a good approximation for a wide range of matter density profiles.

Chapter 4

The solar neutrino problem

4.1 The standard solar model and missing solar neutrinos

The Sun produces energy through nuclear fusion reactions in its core, which also produce a large number of neutrinos. The major reaction is the proton-proton chain, $4p \rightarrow He + 2e^+ + 2\nu_e$, though there are other reactions that produce heavier elements like beryllium and boron and give off ν_e. The various reactions inside the sun producing neutrinos and their energy spectra are predicted by the standard solar model (SSM). The flux v.s. energy spectra of neutrinos produced in various reactions inside the sun, according to the SSM, are shown in 4.1. The electroweak theory predicts the shapes of the spectra, but the absolute magnitudes of the fluxes are calculated from the SSM.

The first indication that something was missing came from the observation of solar neutrinos through radiochemical reactions. Gallium and chlorine were the two nuclei used, which would absorb ν_e and convert to radioactive nuclei with the lifetime of the order of a few weeks. The relevant radiochemical reactions and their threshold energies are:

$$\text{Gallium}: \quad {}^{71}Ga + \nu_e \rightarrow {}^{71}Ge + e^- \quad (233 \text{ keV}) \tag{4.1}$$

$$\text{Chlorine}: \quad {}^{37}Cl + \nu_e \rightarrow {}^{37}Ar + e^- \quad (814 \text{ keV}) \tag{4.2}$$

In addition, in water Cherenkov detectors like Kamiokande (and later Super-Kamiokande), ν_e were detected through the "elastic scattering" processes

$$\nu_e + e^- \rightarrow \nu_e + e^- \tag{4.3}$$

where the electron was detected through its Cherenkov radiation. The threshold for the detection through this process is around 5 MeV, the limiting factor here being the rejection of the background noise in the photomultiplier tubes. The combined observations of these experiments suggested that the flux of ν_e from the sun was smaller than expected by almost a factor of two. Changing the relative flux ratios between ν_e from different reactions also did not help matters.

4.2 Solar neutrino solutions

The neutrinos are produced in the core of the sun where the density is very high, and they have to propagate through the varying density of the sun to reach the surface and then

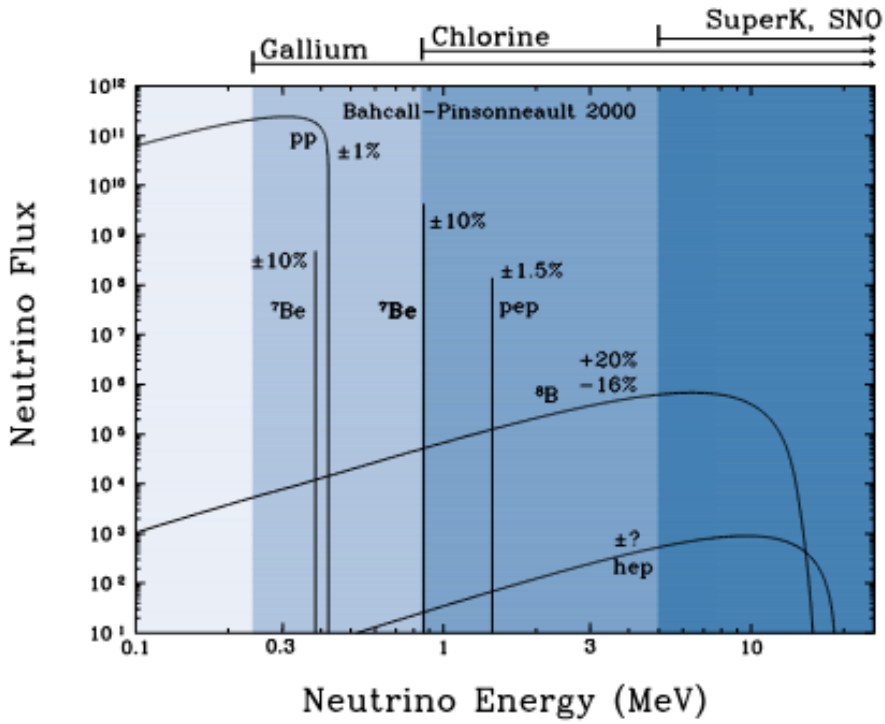

Figure 4.1: The flux v.s. energy spectra of neutrinos produced in various reactions inside the sun

travel through the interstellar space to reach the Earth (which is essentially vacuum). Hence, when they are produced, $A \gg \Delta m^2$, and the effective mixing angle in matter θ_m is close to 90°, which means that the (electron) neutrinos are almost in the heavier mass eigenstate in matter. When they reach earth, they are in vacuum, $A \approx 0$, and the effective mixing angle in matter θ_m is close to the mixing angle in vacuum θ. In vacuum, the electron neutrinos are almost in the lighter mass eigenstate in vacuum. The neutrinos travel through the sun with varying density. Using the level crossing diagram, we can see that the neutrinos will pass through the resonance point where $A = \Delta m^2 \cos 2\theta$. Hence, the probability of detection of the neutrinos as electron neutrinos on Earth would be given by:

$$P_{\nu_e \to \nu_e} = P_{jump} \cos^2\theta + (1 - P_{jump}) \sin^2\theta \tag{4.4}$$

Writing $P_{jump} = e^{-\pi\gamma_{res}/2}$, where $\gamma_{res} = \frac{\Delta m^2 \sin^2 2\theta}{2E\cos 2\theta} \left|\frac{1}{A}\frac{dA}{dx}\right|^{-1}_{res}$, we have

$$P_{\nu_e \to \nu_e} = \exp\Big(-\frac{\pi\Delta m^2 \sin^2 2\theta}{4E\cos 2\theta}\left|\frac{1}{A}\frac{dA}{dx}\right|^{-1}_{res}\Big)\cos^2\theta + \Big(1 - \exp\Big(-\frac{\pi\Delta m^2 \sin^2 2\theta}{4E\cos 2\theta}\left|\frac{1}{A}\frac{dA}{dx}\right|^{-1}_{res}\Big)\Big)\sin^2\theta \tag{4.5}$$

The above formula can be understood as follows: the neutrinos are produced in the heavier mass eigenstate in matter, and if they jump to the lighter mass eigenstate ν_1 in matter with probability P_{jump} as they come out of the sun, then they will be detected as electron neutrinos with probability $\cos^2\theta$ because at earth, $\nu_e = \cos\theta\nu_1 - \sin\theta\nu_2$ (since space between the Sun and the Earth is essentially vacuum, and hence v_{1m} and v_{2m} don't mix with each other and hence the neutrinos will remain in the same mass eigenstate as

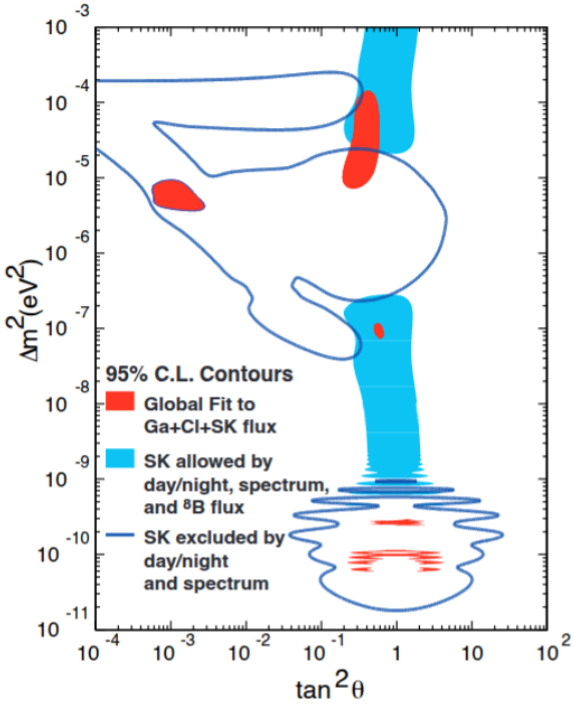

Figure 4.2: The allowed regions for Δm^2 and $\tan^2\theta$ from the solar neutrino experiments.

they propagate through space). On the other hand, if they don't jump to the lighter mass eigenstate in matter with probability $1 - P_{jump}$, then they will be detected as electron neutrinos with probability $\sin^2\theta$. The P_{jump} can be calculated using the Landau-Zener formula as mentioned in equation 3.14.

Using the ν_e flux measurements from the various solar neutrino experiments, the allowed regions for Δm^2 and $\tan^2\theta$ are presented in the figure 4.2. The figure is old (2001), and the allowed regions have shrunk since then, but the main point is that there are four allowed regions. There are two solutions with $\Delta m^2 > 10^{-6}$ eV^2 that correspond to the MSW solutions, where the matter effects in the sun play a significant role The one where $\theta \ll 1$ (and $P_{jump} \approx 1$) is called the small mixing angle (SMA) solution, and the one where $\theta \sim 1$ (and $P_{jump} \approx 0$) is called the large mixing angle (LMA) solution. For both these solutions, the Δm^2 is large enough that there is decoherence between the two mass eigenstates ν_{1m} and ν_{2m}. The solution with $\Delta m^2 \sim 10^{-7}$ eV^2, $\theta \sim 1$, called the LOW solution is another MSW solution, where the matter effects in the sun play a significant role, but the Δm^2 is small enough that there is no decoherence between the two mass eigenstates ν_{1m} and ν_{2m}. The solution with $\Delta m^2 \sim 10^{-10}$ eV^2, $\theta \sim 1$, called the vacuum oscillation (VAC) solution is the one where the matter effects in the sun do not play a significant role (since the resonance condition $A = 2EV_{CC} = \Delta m^2 \cos 2\theta$ is not satisfied for the relevant neutrino energies). Here, the resonance inside the sun is completely non-adiabatic, and the neutrinos are produced in the heavier mass eigenstate in matter, but they come out of the sun as the lighter mass eigenstate in vacuum ($P_{jump} \approx 1$), and the oscillations of the neutrinos take place in vacuum as they travel from the sun to the earth. All except the LMA solutions have been ruled out by the data. The phenomenology here is as follows:

- For the SMA solution, the neutrinos are produced in the heavier mass eigenstate in

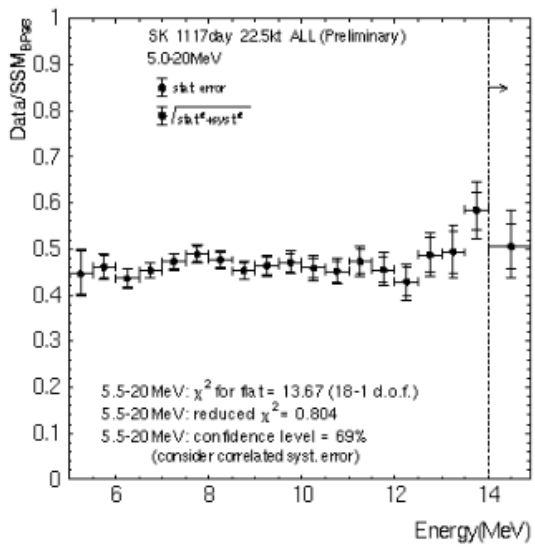

Figure 4.3: The energy dependence of the ν_e survival probability from the solar neutrino data. The data show no energy dependence for $E > 5$ MeV, thus strongly favouring the LMA solution and ruling out the other two.

matter, and they come out of the sun as the lighter mass eigenstate in vacuum with probability $P_{jump} \approx 1$. However, the paramters are such that they undergo MSW resonance as they travel inside the Earth. Hence one would expect a significant day-night asymmetry in the flux of solar neutrinos, which is not observed.

- For the VAC solution, the oscillations of the neutrinos take place in vacuum as they travel from the sun to the earth, and hence one would expect a significant seasonal variation in the flux of solar neutrinos due to the eccentricity of the Earth's orbit around the Sun, which is not observed.

- Both the VAC and the SMA solution imply an energy dependence of the ν_e survival probability: the VAC solution through the $\sin^2(\Delta m^2 L/E)$ term and the SMA solution through the energy dependence of P_{jump} in Landau-Zener formula. The LMA solution gives $P_{jump} = 0$ throughout the energy range $E > 5$ MeV and hence predicts no energy dependence. The data, as shown in Fig.4.3, show no energy dependence for $E > 5$ MeV, thus strongly favouring the LMA solution and ruling out the other two.

Note that the above discussion is just to give physical insight into the solar neutrino problem and its solution through neutrino oscillations. A detailed analysis of the data is done using fit to the including all the parameters. Through this, the LOW solution is also ruled out, and the LMA solution is the only solution that is consistent with the data.

4.3 Confirming the solar neutrino oscillations

The oscillation hypothesis that explains the solar neutrino observations involves the conversion of ν_e into ν_μ, ν_τ or some linear combination of them. However, the experiments

mentioned above are not able to confirm that the ν_μ or ν_τ are indeed produced. That evidence came from the SNO (Sudbury Neutrino Observatory) experiment that used heavy water as the detector, and detected neutrinos through three reactions:

- Elastic scattering (ES) $\nu + e^- \to \nu + e^-$: This reaction detects all three neutrino species, ν_e, ν_μ and ν_τ. However, the cross section of the reaction involving ν_e is almost 6 times the cross section of the reaction involving the other two species, so that this reaction can be said to be sensitive to the combination of fluxes $\Phi_e + \Phi_{\mu+\tau}/6$.
- Charged current (CC) $\nu_e + d \to p + p + e^-$: This reaction can take place only with ν_e, and hence is sensitive to Φ_e.
- Neutral current (NC) $\nu + d \to n + p + \nu$: This reaction is blind to the flavour of the neutrino species and hence effectively measures $\Phi_e + \Phi_{\mu+\tau}$.

In SNO, the NC process was detected through the addition of salt to the heavy water, which was useful in capturing neutrons. The observations confirmed that all the disappearing ν_e were indeed getting converted to ν_μ or ν_τ. This ruled out the conversion of ν_e to a sterile fermion as the mechanism for the solar neutrino oscillations. The neutral current data also matched with the fluxes predicted by SSM, thus vindicating SSM as the correct model of the solar dynamics.

4.4 Confirming the solar neutrino oscillations on earth

With the solar parameters approximately known, it was possible to confirm them using a neutrino source on the earth and observing the depletion of the flux after the neutrinos travel a known distance. The $\bar{\nu_e}$ that are produced in nuclear reactors have energies of a few MeV, and hence with the oscillation parameters obtained by the solar neutrino fit, an oscillation length in vacuum of $\sim$ 100 km. The oscillation parameters can then be tested by having a detector at the appropriate distance from the source. The KamLand experiment, which used the combined flux of many nuclear reactors in Japan and a scintillation detector, confirmed the LMA solution and measured the values of Δm^2 and θ to a much better accuracy. The best fit values for the mixing parameters are $\Delta m^2 = (7.59 \pm 0.21) \times 10^{-5}$ eV^2 and $\tan^2\theta = 0.47^{+0.06}_{-0.05}$[7]. The LMA solution, confirmed by all the experimental data, corresponds to (for $E > 5$ MeV) a completely adiabatic transition ($P_{jump} = 0$) at the resonance the neutrinos undergo inside the sun. Thus, ν_e, which are produced as ν_{2m} inside the sun, travel through the matter inside the sun and emerge from the sun as ν_2. These ν_2, being mass eigenstates in vacuum, reach the earth as ν_2, where they are detected as ν_e with a probability $\sin^2\theta$ and as $\nu_{\mu/\tau}$ with a probability $\cos^2\theta$.

Chapter 5

Three neutrino framework

Since ν_e do not participate in the atmospheric neutrino oscillations, whereas they do participate in the solar neutrino oscillations, it is clear that in order to have a consistent picture of neutrino mixings, we have to develop a framework for the mixing of all three neutrino species, ν_e, ν_μ, ν_τ. We express these flavour eigenstates as linear combinations of the three mass eigenstates in vacuum, ν_1, ν_2, ν_3 as

$$\nu_\alpha = \sum_i U_{\alpha i}\nu_i \tag{5.1}$$

where the mixing matrix U can be parametrized as a product of three rotation matrices:

$$U = R_{23}(\theta_{23})R_{13}(\theta_{13})R_{12}(\theta_{12}) \tag{5.2}$$

We neglect CP violation for the moment, and shall introduce it in the later section. In the absence of CP violation, the 3×3 mixing matrix is real and is completely described in terms of the three mixing angles, θ_{12}, θ_{13}, θ_{23}. It can be explicitly written as

$$U = \begin{pmatrix} 1 & 0 & 0 \\ 0 & c_{23} & s_{23} \\ 0 & -s_{23} & c_{23} \end{pmatrix} \begin{pmatrix} c_{13} & 0 & s_{13} \\ 0 & 1 & 0 \\ -s_{13} & 0 & c_{13} \end{pmatrix} \begin{pmatrix} c_{12} & s_{12} & 0 \\ -s_{12} & c_{12} & 0 \\ 0 & 0 & 1 \end{pmatrix} \tag{5.3}$$

$$= \begin{pmatrix} c_{12}c_{13} & s_{12}c_{13} & s_{13} \\ -s_{12}c_{23} - c_{12}s_{23}s_{13} & c_{12}c_{23} - s_{12}s_{23}s_{13} & s_{23}c_{13} \\ s_{12}s_{23} - c_{12}c_{23}s_{13} & -c_{12}s_{23} - s_{12}c_{23}s_{13} & c_{23}c_{13} \end{pmatrix} \tag{5.4}$$

where $c_i \equiv \cos\theta_i$ and $s_i \equiv \sin\theta_i$.

If a neutrino flavour eigenstate $|\nu\rangle = |\nu_\alpha\rangle$ is produced at time $t = 0$, then the probability that it will be converted to another flavour eigenstate ν_β at a later time t can be calculated as we had seen in the case of two neutrino mixing. Expressing $|\nu_\alpha\rangle$ in terms of the mass eigenstates, we have

$$|\nu_\alpha\rangle = \sum_i U_{\alpha i}|\nu_i\rangle \tag{5.5}$$

The mass eigenstates evolve in time as $|\nu_i(t)\rangle = e^{-iE_it}|\nu_i\rangle$, where E_i is the energy of the mass eigenstate ν_i. Hence, we have

$$|\nu_\alpha(t)\rangle = \sum_i U_{\alpha i}e^{-iE_it}|\nu_i\rangle \tag{5.6}$$

We know $E_i = \sqrt{p^2 + m_i^2} \approx p + \frac{m_i^2}{2E}$, where E is the energy of the neutrino. The common phase e^{-ipt} can be factored out, and we have

$$|\nu_\alpha(t)\rangle = e^{-ipt} \sum_i U_{\alpha i} e^{-i\frac{m_i^2}{2E}t} |\nu_i\rangle \tag{5.7}$$

The probability of conversion to another flavour eigenstate ν_β is $|\langle\nu_\beta|\nu_\alpha(t)\rangle|^2$. Expressing $|\nu_\beta\rangle$ in terms of the mass eigenstates, we have

$$\begin{aligned}
|\nu_\beta\rangle &= \sum_j U_{\beta j} |\nu_j\rangle \\
\Rightarrow \langle\nu_\beta| &= \sum_j U^*_{\beta j} \langle\nu_j| \\
\Rightarrow \langle\nu_\beta|\nu_\alpha(t)\rangle &= e^{-ipt} \sum_{i,j} U^*_{\beta j} U_{\alpha i} e^{-i\frac{m_i^2}{2E}t} \langle\nu_j|\nu_i\rangle \\
&= e^{-ipt} \sum_i U^*_{\beta i} U_{\alpha i} e^{-i\frac{m_i^2}{2E}t}
\end{aligned}$$

Hence, the probability of conversion is given by

$$\begin{aligned}
P_{\alpha\beta} &= |\langle\nu_\beta|\nu_\alpha(t)\rangle|^2 \\
&= \left| e^{-ipt} \sum_i U^*_{\beta i} U_{\alpha i} e^{-i\frac{m_i^2}{2E}t} \right|^2 \\
&= \sum_{i,j} U^*_{\beta j} U_{\alpha j} U_{\beta i} U^*_{\alpha i} e^{-i\frac{m_i^2}{2E}t} e^{i\frac{m_j^2}{2E}t} \\
&= \sum_{i,j} U^*_{\beta j} U_{\alpha j} U_{\beta i} U^*_{\alpha i} e^{-i\frac{\Delta m_{ij}^2}{2E}t} \\
&= \sum_{i=j} U^*_{\beta j} U_{\alpha j} U_{\beta i} U^*_{\alpha i} + \sum_{i>j} U^*_{\beta j} U_{\alpha j} U_{\beta i} U^*_{\alpha i} e^{-i\frac{\Delta m_{ij}^2}{2E}t} + \sum_{i<j} U^*_{\beta j} U_{\alpha j} U_{\beta i} U^*_{\alpha i} e^{-i\frac{\Delta m_{ij}^2}{2E}t} \\
&= \sum_i |U_{\beta i}|^2 |U_{\alpha i}|^2 + \sum_{i>j} U^*_{\beta j} U_{\alpha j} U_{\beta i} U^*_{\alpha i} e^{-i\frac{\Delta m_{ij}^2}{2E}t} + \sum_{i>j} U^*_{\beta i} U_{\alpha i} U_{\beta j} U^*_{\alpha j} e^{i\frac{\Delta m_{ij}^2}{2E}t} \\
&= \sum_i |U_{\beta i}|^2 |U_{\alpha i}|^2 + \sum_{i>j} U^*_{\beta j} U_{\alpha j} U_{\beta i} U^*_{\alpha i} \cos\left(\frac{\Delta m_{ij}^2 L}{2E}\right) - i \sum_{i>j} U^*_{\beta j} U_{\alpha j} U_{\beta i} U^*_{\alpha i} \sin\left(\frac{\Delta m_{ij}^2 L}{2E}\right) \\
&+ \sum_{i>j} U^*_{\beta i} U_{\alpha i} U_{\beta j} U^*_{\alpha j} \cos\left(\frac{\Delta m_{ij}^2 L}{2E}\right) + i \sum_{i>j} U^*_{\beta i} U_{\alpha i} U_{\beta j} U^*_{\alpha j} \sin\left(\frac{\Delta m_{ij}^2 L}{2E}\right) \\
&= \sum_i |U_{\beta i}|^2 |U_{\alpha i}|^2 + 2 \sum_{i>j} \mathrm{Re}(U^*_{\beta j} U_{\alpha j} U_{\beta i} U^*_{\alpha i}) \cos\left(\frac{\Delta m_{ij}^2 L}{2E}\right) + 2 \sum_{i>j} \mathrm{Im}(U^*_{\beta j} U_{\alpha j} U_{\beta i} U^*_{\alpha i}) \sin\left(\frac{\Delta m_{ij}^2 L}{2E}\right) \\
&= \delta_{\alpha\beta} - 4 \sum_{i>j} \mathrm{Re}(U^*_{\beta j} U_{\alpha j} U_{\beta i} U^*_{\alpha i}) \sin^2\left(\frac{\Delta m_{ij}^2 L}{4E}\right) + 2 \sum_{i>j} \mathrm{Im}(U^*_{\beta j} U_{\alpha j} U_{\beta i} U^*_{\alpha i}) \sin\left(\frac{\Delta m_{ij}^2 L}{2E}\right)
\end{aligned}$$

Here we have used $\cos x = 1 - 2\sin^2(x/2)$ in the last step along with the unitarity of

the mixing matrix U as follows:

$$\begin{aligned}
\sum_i |U_{\beta i}|^2 |U_{\alpha i}|^2 + 2\sum_{i>j} \mathrm{Re}(U^*_{\beta j} U_{\alpha j} U_{\beta i} U^*_{\alpha i}) &= \sum_{i=j} U^*_{\beta j} U_{\alpha j} U_{\beta i} U^*_{\alpha i} + \sum_{i>j} (U^*_{\beta j} U_{\alpha j} U_{\beta i} U^*_{\alpha i} + U^*_{\beta i} U_{\alpha i} U_{\beta j} U^*_{\alpha j}) \\
&= \sum_{i=j} U^*_{\beta j} U_{\alpha j} U_{\beta i} U^*_{\alpha i} + \sum_{i \neq j} U^*_{\beta j} U_{\alpha j} U_{\beta i} U^*_{\alpha i} \\
&= \sum_{i,j} U^*_{\beta j} U_{\alpha j} U_{\beta i} U^*_{\alpha i} \\
&= \left(\sum_j U^*_{\beta j} U_{\alpha j} \right) \left(\sum_i U_{\beta i} U^*_{\alpha i} \right) \\
&= \delta_{\alpha\beta} \delta_{\beta\alpha} \\
&= \delta_{\alpha\beta}
\end{aligned}$$

If the CP violation is neglected, then all the terms in the matrix are real, and the last term in the expression for $P_{\alpha\beta}$ vanishes, and we have

$$P_{\alpha\beta} = \delta_{\alpha\beta} - 4 \sum_{i>j} U_{\beta j} U_{\alpha j} U_{\beta i} U_{\alpha i} \sin^2 \left(\frac{\Delta m^2_{ij} L}{4E} \right) \tag{5.8}$$

and hence the survival probability of a neutrino flavour eigenstate ν_α is given by

$$P_{\alpha\alpha} = 1 - 4 \sum_{i>j} |U_{\alpha j}|^2 |U_{\alpha i}|^2 \sin^2 \left(\frac{\Delta m^2_{ij} L}{4E} \right) \tag{5.9}$$

5.1 Measurement of θ_{13}

We already know from the solar neutrino and atmospheric neutrino data that $|\Delta m^2_\odot|$ ($\sim 10^{-5}$ eV^2) is much smaller than $|\Delta m^2_{atm}|$ ($\sim 10^{-3}$ eV^2). Hence the mass spectrum of neutrinos consist of two species that are close to each other in mass $\Delta m^2_\odot$, and a third species that is separated from the other two by Δm^2_{atm}. We define the mass eigenstate ν_3 to be the one that is separated from the other two by Δm^2_{atm}, and the mass eigenstates ν_1 and ν_2 to be the two that are close to each other in mass. Hence, we have $\Delta m^2_{21} = \Delta m^2_\odot$ and $\Delta m^2_{31} \approx \Delta m^2_{32} = \Delta m^2_{atm}$. Hence, the survival probability of ν_e in the three neutrino framework can be written as

$$P_{ee} = 1 - 4|U_{e1}|^2 |U_{e2}|^2 \sin^2 \left(\frac{\Delta m^2_{21} L}{4E} \right) - 4|U_{e1}|^2 |U_{e3}|^2 \sin^2 \left(\frac{\Delta m^2_{31} L}{4E} \right) \tag{5.10}$$

$$- 4|U_{e2}|^2 |U_{e3}|^2 \sin^2 \left(\frac{\Delta m^2_{32} L}{4E} \right) \tag{5.11}$$

$$\approx 1 - 4||U_{e3}|^2 (1 - |U_{e3}|^2) \sin^2 \left(\frac{\Delta m^2_{31} L}{4E} \right) \tag{5.12}$$

We have used the fact that $\Delta m^2_{21} \ll \Delta m^2_{31} \approx \Delta m^2_{32}$, and hence the oscillation term involving Δm^2_{21} can be neglected (since the experiment was designed with $\Delta m^2_{21} L/E \ll 1$). We have also used the unitarity of the mixing matrix to write $|U_{e1}|^2 + |U_{e2}|^2 + |U_{e3}|^2 = 1$

to write the last two terms in the expression for P_{ee} in terms of $|U_{e3}|^2$. Recall from equation 5.4 that $|U_{e3}|^2 = \sin^2\theta_{13}$. Hence, we have

$$P_{ee} \approx 1 - \sin^2(2\theta_{13})\sin^2\left(\frac{\Delta m^2_{31}L}{4E}\right) \tag{5.13}$$

The CHOOZ experiment[8], which was a reactor neutrino experiment with a baseline of about 1 km, did not observe any disappearance of $\bar{\nu_e}$, and hence put an upper bound on θ_{13} as $\sin^2(2\theta_{13}) < 0.05$ at 3σ confidence level.

5.2 Solar and atmospheric neutrino oscillations in the three neutrino framework

We already know that $\Delta m^2_{\odot} \ll \Delta m^2_{atm}$. The neutrino mass spectrum thus consists of two mass eigenstates relatively close together (separated by $\Delta m^2_{\odot}$) and one mass eigenstate comparatively further away from these two. We define the eigenstate further away by ν_3. Out of the remaining two eigenstates, the heavier one is ν_2 and the lighter one ν_1, by definition. The atmospheric neutrinos measure the survival probability of ν_μ in vacuum, i.e.

$$P_{\mu\mu} = 1 - 4|U_{\mu1}|^2|U_{\mu2}|^2\sin^2\left(\frac{\Delta m^2_{21}L}{4E}\right) - 4|U_{\mu2}|^2|U_{\mu3}|^2\sin^2\left(\frac{\Delta m^2_{32}L}{4E}\right) \tag{5.14}$$

$$- 4|U_{\mu1}|^2|U_{\mu3}|^2\sin^2\left(\frac{\Delta m^2_{31}L}{4E}\right) \tag{5.15}$$

$$\approx 1 - 4|U_{\mu3}|^2(1-|U_{\mu3}|^2)\sin^2\left(\frac{\Delta m^2_{32}L}{4E}\right) \tag{5.16}$$

where we have neglected the term containing $\sin^2\left(\frac{\Delta m^2_{21}L}{4E}\right)$ because of its smallness, used $\Delta m^2_{32} \approx \Delta m^2_{31}$, and used the unitarity of the mixing matrix:

$$|U_{\mu1}|^2 + |U_{\mu2}|^2 + |U_{\mu3}|^2 = 1$$

Since $\cos\theta_{13} \approx 1$, we have $4|U_{\mu3}|^2(1-|U_{\mu3}|^2) \approx \sin^2(2\theta_{23})$. Comparing the survival probability with the one obtained in eq.(2.10) using two neutrino mixing, we get

$$\theta_{atm} \approx \theta_{23} \tag{5.17}$$

so that the atmospheric neutrino observations give

$$\Delta m^2_{atm} \approx |\Delta m^2_{31}| \approx 2\times10^{-3}\ \text{eV}^2, \theta_{atm} \approx \theta_{23} \approx 45^\circ \tag{5.18}$$

Note that only the magnitude of Δm^2_{31} is measured in the atmospheric neutrino experiments, and not its sign. As a result, there are two possible mass hierarchies in neutrinos, one "normal" where $m_1 < m_2 < m_3$ and the other "inverted" where $m_3 < m_1 < m_2$. The hierarchies are shown in Fig. 5.1. In the case of solar neutrino solution, using $\theta_{13} \approx 0$ one can show that

$$\Delta m^2_{\odot} \approx \Delta m^2_{21} \approx 8\times10^{-5}\ \text{eV}^2, \theta_{\odot} \approx \theta_{12} \approx 32^\circ \tag{5.19}$$

Thus, due to the smallness of θ_{13} and the "hierarchy" in Δm^2, the solutions to the solar and atmospheric neutrino anomalies are completely decoupled, and have a simple correspondence with the angles that parametrize the mixing matrix. The measurement of θ_{13} and determining the sign of Δm^2_{31} are two of the important future challenges for neutrino experiments.

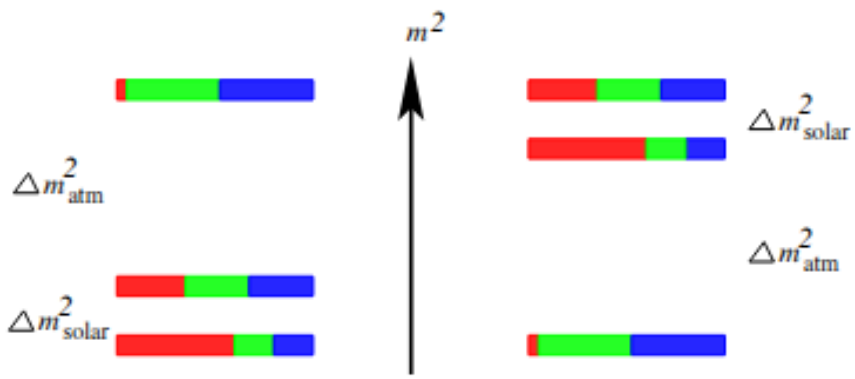

Figure 5.1: The two possible mass hierarchies in neutrinos: the normal hierarchy where $m_1 < m_2 < m_3$ and the inverted hierarchy where $m_3 < m_1 < m_2$. The red, green and blue colors represent the composition of the ν_e, ν_μ and ν_τ flavour eigenstates in the mass eigenstates.

Chapter 6

Detection of Neutrinos

Neutrinos are notoriously difficult to detect due to their weak interactions with matter. Hence, the detection of neutrinos requires large detectors and intense sources of neutrinos. The detection of neutrinos is based on the observation of the products of their interactions with matter. The most common interactions used for neutrino detection are:

- Charged current (CC) interactions: $\nu_\alpha + N \to \alpha + X$, where N is a nucleon and X is the hadronic final state. In this interaction, the neutrino converts into its corresponding charged lepton, which can be detected through its ionization or Cherenkov radiation.
- Neutral current (NC) interactions: $\nu_\alpha + N \to \nu_\alpha + X$, where the neutrino scatters off a nucleon without changing its flavour. The hadronic final state can be detected through its ionization or Cherenkov radiation.
- Elastic scattering (ES): $\nu_\alpha + e^- \to \nu_\alpha + e^-$, where the neutrino scatters off an electron. The scattered electron can be detected through its ionization or Cherenkov radiation.

Atmospheric and solar neutrinos are discussed in the previous sections. Apart from these, there are also neutrinos produced in astrophysical sources such as supernovae, active galactic nuclei, and gamma-ray bursts. These neutrinos have much higher energies than the atmospheric and solar neutrinos. The crucial point is that, unlike a photon or a cosmic-ray particle, neutrinos can travel through the universe without being absorbed or deflected by magnetic fields, and hence they can provide us with information about the processes happening in these astrophysical sources. The detection of these high-energy neutrinos is done using large volume detectors such as IceCube, which is a cubic kilometer of ice at the South Pole instrumented with photomultiplier tubes to detect the Cherenkov radiation produced by the charged particles produced in the interactions of neutrinos with the ice.

6.1 Basics of Cherenkov radiation

When a charged particle travels through a medium with a speed greater than the speed of light in that medium, it emits Cherenkov radiation. The speed of light in a medium with refractive index n is given by c/n, where c is the speed of light in vacuum. Hence, the condition for Cherenkov radiation to be emitted is $v > c/n$, where v is the speed of

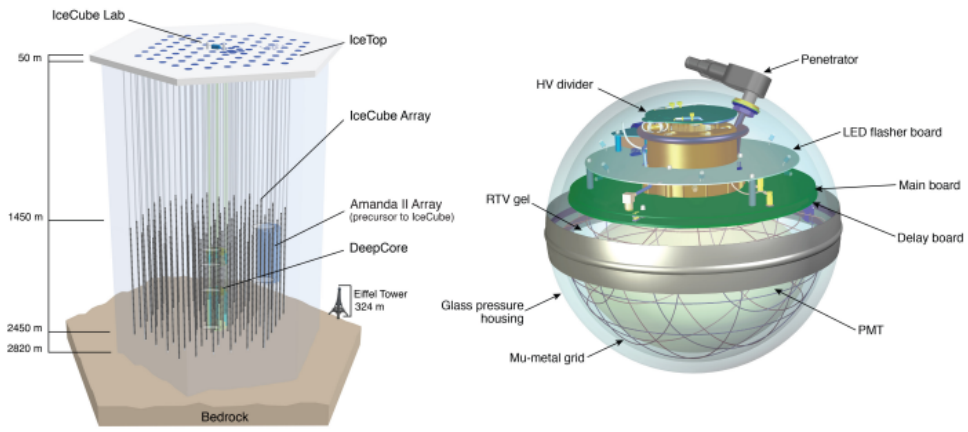

Figure 6.1: Sketch of the IceCube observatory (left) and the digital optical module (right)

the charged particle. The Cherenkov radiation is emitted in a cone with an opening angle θ_c given by $\cos\theta_c = \frac{1}{\beta n}$, where $\beta = v/c$, and n is the refractive index of the medium. The intensity of the Cherenkov radiation is proportional to $\sin^2\theta_c$, and hence it increases as the speed of the charged particle increases. The Cherenkov radiation can be detected using photomultiplier tubes, which convert the Cherenkov radiation into an electrical signal that can be recorded and analyzed.

6.2 Detection of high-energy neutrinos

High-energy neutrinos interact predominantly with matter via deep inelastic scattering off nucleons: the neutrino scatters off quarks in the target nucleus by the exchange of a Z or W weak boson, referred to as neutral current (NC) and charged current (CC) interactions, respectively. Whereas the NC interaction leaves the neutrino flavour unchanged, the CC interaction converts the neutrino into its corresponding charged lepton. A variety of methods are used to detect the high-energy secondary particles created in CC and NC neutrino interactions. One particularly effective method observes the radiation of optical Cherenkov light radiated by secondary charged particles produced in CC and NC interactions that travel faster than the speed of light in the medium.

The detection concept is that of a conventional Cherenkov detector, a transparent medium is instrumented with photo-multipliers that transform the Cherenkov light into electrical signals by the photoelectric effect; see Figs. 6.1. IceCube consists of 80 strings, each instrumented with 60 10-inch photo-multipliers spaced by 17 m over a total length of 1 kilometer. The deepest module is located at a depth of 2.450 km so that the instrument is shielded from the large background of cosmic rays at the surface by approximately 1.5 km of ice. Strings are arranged at apexes of equilateral triangles that are 125 m on a side. The instrumented detector volume is a cubic kilometer of dark, highly transparent and sterile Antarctic ice. The radioactive background in the detector is dominated by the instrumentation deployed into this natural ice.[1]

Each optical sensor consists of a glass sphere containing the photomultiplier and the electronics board that captures and digitizes the signals locally using an onboard computer. The digitized signals are given a global time stamp with residuals accurate to

[1]Source for this section: Neutrino review by F. Hanzel et.al [9]

less than 3 ns and are subsequently transmitted to the surface. Processors at the surface continuously collect the time-stamped signals from the optical modules, each of which functions independently. The digital messages are sent to a string processor and a global event trigger. They are subsequently sorted into the Cherenkov patterns emitted by secondary muon tracks, or particle showers for electron and tau neutrinos, that reveal the flavor, energy and direction of the incident neutrino.58 There are two principle classes of Cherenkov events that must be separated by the detector, "tracks" and "cascades". The two basic topologies are illustrated in Fig. 6.2: tracks initiated by ν_μ neutrinos and cascades from ν_e, ν_τ neutrinos as well as the neutral current interactions from all flavors. On the scale of IceCube, PeV cascades, with a length of less than 10 m, are therefore essentially point sources of Cherenkov light in a detector of kilometer size. The term "tracks" refers to the Cherenkov emission of long-lived muons produced in CC interactions of muon neutrinos interacting inside or in the vicinity of the detector. Energetic electrons and taus produced in CC interactions of electron and tau neutrino interactions, respectively, will not produce elongated tracks: electrons will initiate electromagnetic showers in the ice and, because of the relatively short lifetime, taus will decay and also produce a shower in the ice. Because of the large background of muons produced by cosmic ray interactions in the atmosphere, the observation of muon neutrinos is typically limited to upgoing muon tracks that are produced in interactions inside or close to the detector by neutrinos that have passed through the Earth as illustrated in Fig. 6.3. The remaining background consists of atmospheric neutrinos, which are indistinguishable from cosmic neutrinos on an event-by-event basis. However, the steeply falling spectrum ($\propto E^{-3.7}$) of atmospheric neutrinos allows identifying diffuse astrophysical neutrino by a spectral analysis as we will highlight in the following sections. Above $\sim$ 300 TeV atmospheric neutrinos are relatively rare, even in a cubic kilometer detector, and every neutrino exceeding this energy is likely to be of cosmic origin. The atmospheric background is further reduced when looking in the specific direction of a point source, in particular transient neutrino sources like GRB. The hadronic particle shower generated by the target struck by a neutrino also radiates Cherenkov photons. Because of the large multiplicity of secondary particles at these energies and the repeated scattering of the Cherenkov photons in the ice, the light pattern is essential spherical tens of meters from its point of origin. The light patterns produced by the particle showers initiated by the electron or tau produced in CC interactions of electron or tau neutrinos, respectively, will be superimposed on the hadronic cascade. The two are not typically not separated. The direction of the initial neutrino can only be reconstructed from the Cherenkov emission of secondary particles produced close to the neutrino interaction point, and the angular resolution is inferior to that for track events. In contrast, the energy resolution of the neutrino is superior for cascades than for tracks. For both, the observable energy of the secondaries can be estimated from the total number of Cherenkov photons after accounting for kinematic effects and detection efficiencies. The Cherenkov light observed in cascades is proportional to the energy transferred to the cascade and is often fully contained in the instrumented volume. It is actually sufficient that the cascade is partially contained, as long as one can reconstruct its actual size. In contrast, muons produced by CC muon neutrino interactions lose energy gradually by ionization, bremsstrahlung, pair production, and photo-nuclear interactions while passing through the detector. The secondary charged particles produced in each energy-loss interaction radiate Cherenkov photons that allow for a measurement of the total energy lost by the muon in the detector. The energy deposited by the muon in the detector represents a lower limit on the initial neutrino energy. Catastrophically losing energy by

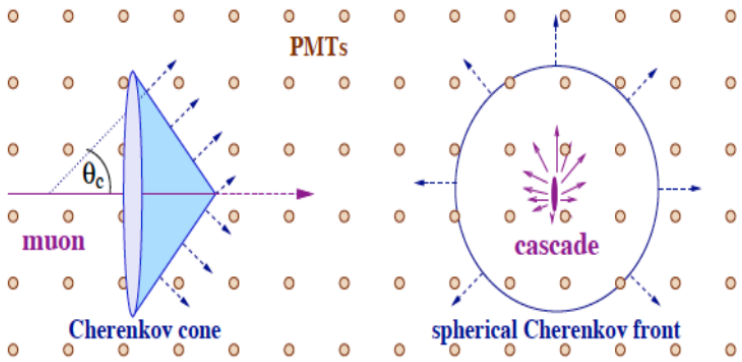

Figure 6.2: The two basic topologies of Cherenkov events in IceCube: tracks initiated by ν_μ neutrinos and cascades from ν_e, ν_τ neutrinos as well as the neutral current interactions from all flavors.

the process mentioned above, muon tracks range out, over kilometers at TeV energy to tens of kilometers at EeV energy. Because the energy of the muon thus degrades along its track, the energy of the secondary showers decreases, which reduces the distance from the track over which the associated Cherenkov light can trigger a PMT. The geometry of the light pool surrounding the muon track is therefore a kilometer-long cone with a gradually decreasing radius. On average, in its first kilometer, a high-energy muon loses energy in a couple of showers with one-tenth of the muon's initial energy. So the initial radius of the cone is the radius of a shower with 10% of the muon energy. At lower energies of hundreds of GeV and less, the muon becomes minimum-ionizing. Because of the stochastic nature of the muon's energy loss, the relationship between the observed energy loss inside the detector and the muon energy varies from muon to muon. Additionally, only the muon energy lost in the detector can be determined; we do not know the energy lost before entering the instrumented volume, nor how much energy it carries out upon exiting. An unfolding process is required to derive the neutrino energy from the energy lost inside the detector; fortunately, it is described by well-understood Standard Model physics. One derives a probability distribution for the energy of the initial neutrino that determines its most probable value. The neutrino energy may only be determined within a factor of 2 or thereabout, depending on the energy, but the uncertainties drop out when measuring a neutrino spectrum involving multiple events. In contrast, for ν_e and ν_τ, the detector is a total energy calorimeter capturing all or most of the Cherenkov light produced, and the determination of their energy is superior. The different topologies each have advantages and disadvantages. For ν_μ charged-current interactions, the long lever arm of muon tracks allows for a measurement of the muon direction with an angular resolution of better than 0.4°. Superior angular resolution can be reached for selected high-energy events. At the highest energies the neutrino is aligned with the muon within the angular resolution and the sensitivity to point sources thus maximized. The disadvantages are a large background of atmospheric neutrinos below 100 TeV, and of cosmic-ray muons at all energies, and the indirect determination of the neutrino energy that must be inferred from sampling the energy loss of the muon when it transits the detector. Observation of ν_e and ν_τ flavors represents significant advantages. They are detected from both Northern and Southern Hemispheres. (This is also true for ν_μ with energy in excess of several hundred TeV, where the background from the steeply falling

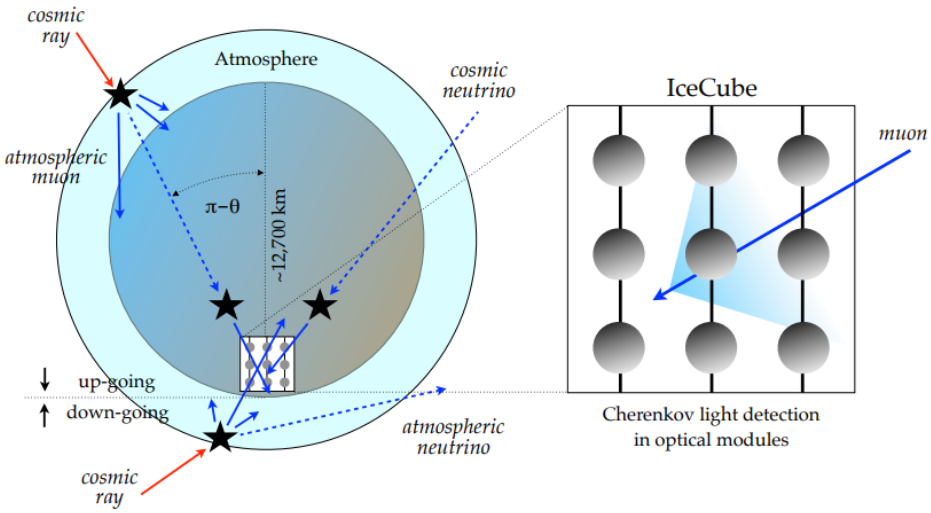

Figure 6.3: The concept of a neutrino telescope from the point of view of IceCube located at the South Pole. Muon neutrinos interact with the nuclei of atoms in a transparent medium like water or ice and produce a secondary muon track that is detected by the wake of Cherenkov photons it leaves inside the detector. The background of high-energy muons (solid blue arrows) produced in the atmosphere can be reduced by placing the detector underground. The surviving fraction of muons is further reduced by looking for upgoing muon tracks that originate from muon neutrinos that have traversed the Earth (dashed blue arrows) interacting close to the detector. This still leaves the contribution of muons generated by atmospheric muon neutrino interactions. This contribution can be separated from the diffuse cosmic neutrino emission by an analysis of the combined neutrino spectrum.

atmospheric spectrum becomes negligible.) At TeV energies and above, the background of atmospheric ν_e is lower by over an order of magnitude because longlived pions, the source of atmospheric ν_e, no longer decay, and relatively rare K decays become the dominant source of background ν_e. Atmospheric ν_τ , produced by oscillations, are rare above an energy of 100 GeV. High-energy ν_τ are therefore of cosmic origin; one such event with an energy of 100 TeV has been identified and represents an independent discovery of cosmic neutrinos. Furthermore, one can establish the cosmic origin of a single single cascade event by demonstrating that the energy cannot be reached by muons and neutrinos of atmospheric origin. Finally, ν_τ are not absorbed by the Earth: ν_τ interacting in the Earth produce a secondary ν_τ of lower energy, either directly in a neutral current interaction or via the decay of a secondary tau lepton produced in a charged-current interaction. High-energy ν_τ will thus cascade down to energies of hundred of TeV where the Earth becomes transparent. In other words, they are detected with a reduced energy, but not absorbed. By this mechanism GZK ν_τ with EeV energies produce a signal at PeV energy. Although cascades are nearly point-like sources of Cherenkov light and, in practice, spatially isotropic, the pattern of arrival times of the photons at individual optical modules reveals the direction of the secondary lepton. While a fraction of cascade events may be reconstructed to within a degree, the precision is inferior to that reached for ν_μ events, typically 8° using the present techniques. A campaign is underway to better characterize the optical properties of the ice and the calibration of the detector in order to improve this resolution. At energies above about 100 PeV, electromagnetic showers begin to elongate because of the Landau-Pomeranchuk-Migdal effect. An extended length scale, associated with the abundant radiation of soft photons, results in the interaction of the secondary shower particles with two atoms. Negative interference of this process relative to interactions with a single atom results in a reduction of the energy loss.

Chapter 7

Dirac versus Majorana Neutrinos

7.1 The Chiral Nature of Mass and the Standard Model

In the formulation of the Standard Model (SM) as a chiral gauge theory, the left-handed neutrinos ν_L are part of the $SU(2)_L$ doublet $L = (\nu_L, \ell_L)^T$, transforming as $(\mathbf{1}, \mathbf{2}, -1/2)$ under the $SU(3)_C \times SU(2)_L \times U(1)_Y$ gauge group. However, there is no corresponding right-handed neutrino field ν_R in the original SM particle content. In quantum field theory, the only renormalizable mass term for fermions is the Dirac mass, which is a Lorentz scalar that fundamentally requires the coupling of a left-handed chiral field to a right-handed chiral field [10]. Therefore, in the minimal SM, neutrinos are strictly massless.

The observation of neutrino oscillations forces us to extend the SM to accommodate non-zero masses. Because neutrinos are the only fermions with zero electric charge and zero color charge, this extension allows for two fundamentally different mathematical pathways: we can introduce a new, sterile right-handed field (Dirac mass), or we can construct a mass term using only the existing left-handed field (Majorana mass). The nature of the mass term determines the underlying degrees of freedom and dictates whether total lepton number is conserved or violated.

7.2 Dirac Neutrinos and Lepton Number Conservation

To give neutrinos a Dirac mass, we extend the SM by introducing a right-handed neutrino field ν_R, which transforms as a singlet $(\mathbf{1}, \mathbf{1}, 0)$ under the SM gauge group. Because it carries no SM gauge charges, it does not interact with the strong, weak, or electromagnetic forces. The allowed Yukawa interaction term is

$$\mathcal{L}_{\text{Yuk}} = -y_\nu \, \bar{L}\tilde{\phi}\nu_R + \text{h.c.}, \tag{7.1}$$

where $L = (\nu_L, \ell_L)^T$ is the lepton doublet, $\tilde{\phi} = i\tau_2\phi^*$ is the conjugate Higgs doublet, and y_ν is a dimensionless Yukawa coupling. After electroweak symmetry breaking, the Higgs acquires a vacuum expectation value $\langle\phi\rangle = (0, v/\sqrt{2})^T$, yielding the Dirac mass term:

$$\mathcal{L}^D_{\text{mass}} = -m_D \left(\overline{\nu_L}\nu_R + \overline{\nu_R}\nu_L\right), \qquad m_D = y_\nu \frac{v}{\sqrt{2}}. \tag{7.2}$$

This mechanism is exactly analogous to the mass generation of charged leptons and quarks. A Dirac neutrino possesses four distinct degrees of freedom: a left-handed and right-handed neutrino (ν_L, ν_R), and their corresponding right-handed and left-handed antineutrinos $(\bar{\nu}_R, \bar{\nu}_L)$.

The vertex factor for this mass insertion is $-im_D$ [11]. The amplitude for a left-handed neutrino to convert into a right-handed one during propagation is given by:

$$i\mathcal{M}_{\text{Dirac}} = \bar{u}_R(-im_D)u_L = -im_D\,\bar{u}_R u_L. \tag{7.3}$$

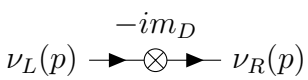

Figure 7.1: Feynman diagram for the Dirac neutrino mass insertion. The incoming left-handed neutrino ν_L is converted into an outgoing right-handed neutrino ν_R via interaction with the Higgs vacuum expectation value $\langle\phi\rangle$. The interaction is denoted by a cross $(\times)$ insertion, corresponding to the vertex factor $-im_D$ [11].

In the Dirac scenario, the Lagrangian is invariant under a global $U(1)_L$ phase transformation $\nu_L \to e^{i\alpha}\nu_L$ and $\nu_R \to e^{i\alpha}\nu_R$. This continuous symmetry corresponds to the absolute conservation of total lepton number L (assigning $L(\nu_R) = +1$). However, the Dirac hypothesis suffers from a severe "naturalness" problem: to yield a sub-eV neutrino mass ($m_D \lesssim 0.1$ eV) given the electroweak scale $v \approx 246$ GeV, the Yukawa coupling must be unnaturally small ($y_\nu \sim 10^{-12}$).

7.3 Majorana Neutrinos, Spinor Algebra, and $\Delta L = 2$

Alternatively, a mass term can be constructed without introducing new light degrees of freedom by utilizing only the left-handed fields. In 1937, Ettore Majorana demonstrated that a neutral fermion can be its own antiparticle. We can construct a right-handed chiral state entirely from the existing ν_L field using the charge conjugation matrix $C = i\gamma^2\gamma^0$, which satisfies $C\gamma^\mu C^{-1} = -(\gamma^\mu)^T$ and $C^T = -C$.

Defining the charge-conjugate field as $\nu_L^c \equiv C\overline{\nu_L}^T$, we note that it transforms as a right-handed spinor under the Lorentz group. We can therefore couple it directly to ν_L to form a Lorentz scalar [10, 12]:

$$\mathcal{L}_{\text{Maj}} = -\frac{1}{2}m_M\left(\overline{\nu_L^c}\nu_L + \overline{\nu_L}\nu_L^c\right) = -\frac{1}{2}m_M\,\nu_L^T C^{-1}\nu_L + \text{h.c.} \tag{7.4}$$

To prove this term does not trivially vanish, we must account for the anti-commuting nature of Grassmann variables. Because fermion fields anti-commute ($\nu_\alpha\nu_\beta = -\nu_\beta\nu_\alpha$), and the matrix C^{-1} is antisymmetric, the transpose of the bilinear is evaluated as [10]:

$$(\nu_L^T C^{-1}\nu_L)^T = -\nu_L^T(C^{-1})^T\nu_L = \nu_L^T C^{-1}\nu_L. \tag{7.5}$$

This confirms the mass term is symmetric and non-vanishing. A Majorana neutrino possesses only two degrees of freedom: the two helicity states of a single self-conjugate particle.

Because the Majorana mass term couples two left-handed neutrino fields, it explicitly violates total lepton number by two units ($\Delta L = 2$). Under a global $U(1)_L$ phase transformation, the bilinear transforms as:

$$\nu_L^T C^{-1} \nu_L \to \left(e^{i\alpha}\nu_L\right)^T C^{-1} \left(e^{i\alpha}\nu_L\right) = e^{2i\alpha} \nu_L^T C^{-1} \nu_L \tag{7.6}$$

meaning the continuous symmetry is explicitly broken down to a discrete $\mathbb{Z}_2$ subgroup, equivalent to $(-1)^L$ conservation.

Furthermore, because the lepton doublet L carries weak isospin $I_3 = 1/2$, the bilinear $\nu_L^T C^{-1} \nu_L$ has weak isospin $I_3 = 1$. Consequently, in the Standard Model, such a term cannot arise from renormalizable interactions with the $I_3 = -1/2$ Higgs doublet. It must instead be generated as an effective operator from new physics at a high scale Λ. The simplest UV completion is the Type I seesaw mechanism [13, 14], which yields the characteristic suppression $m_M \sim y_\nu^2 v^2 / M_R$.

The effective vertex factor for the Majorana mass insertion is $-im_M$. The amplitude for this interaction is:

$$-i\mathcal{M}_{\text{Maj}} = -im_M \, \bar{\nu}_L^c \nu_L. \tag{7.7}$$

This amplitude explicitly carries $\Delta L = 2$ because it annihilates two left-handed neutrinos (or creates two), acting as a sink or source for lepton number.

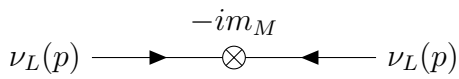

Figure 7.2: Feynman diagram for the effective Majorana neutrino mass insertion. The effective vertex factor is $-im_M$. The clashing fermion arrows explicitly demonstrate the violation of lepton number by two units ($\Delta L = 2$), as the vertex annihilates (or creates) two left-handed neutrinos. In the context of the seesaw mechanism, $m_M \sim y_\nu^2 v^2 / M_R$ [13, 14].

7.4 Neutrinoless Double-Beta Decay ($0\nu\beta\beta$)

Two-Neutrino versus Neutrinoless Decay

To introduce neutrinoless double-beta decay, we must first consider the standard two-neutrino double-beta decay ($2\nu\beta\beta$) processes. A $2\nu\beta\beta^-$ process consists of the simultaneous β^- decay of two neutrons in the same nucleus:

$$\mathcal{N}(A, Z) \to \mathcal{N}(A, Z+2) + 2e^- + 2\bar{\nu}_e \tag{7.8}$$

This process is generated at second order in the perturbative expansion of weak interactions within the Standard Model (SM).

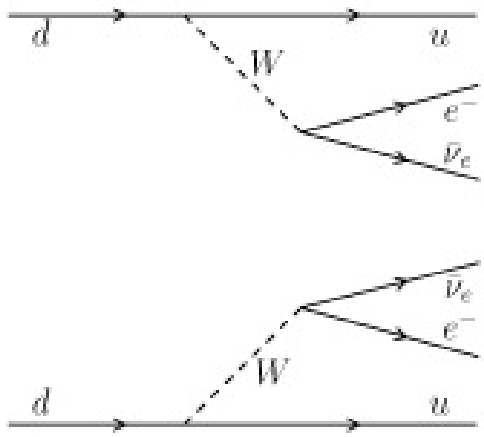

Figure 7.3: Tree-level quark diagram of a $2\nu\beta\beta^-$ process. Adapted from Giunti & Kim [12].

In practice, the 2β-decay of a nucleus is observable only if its single β-decay is energetically forbidden or strongly suppressed because of a large change of spin. For example, in the decay of ^{76}Ge $\rightarrow$ ^{76}Se, the intermediate ^{76}As state has a higher ground-state energy, making the single beta decay kinematically impossible.

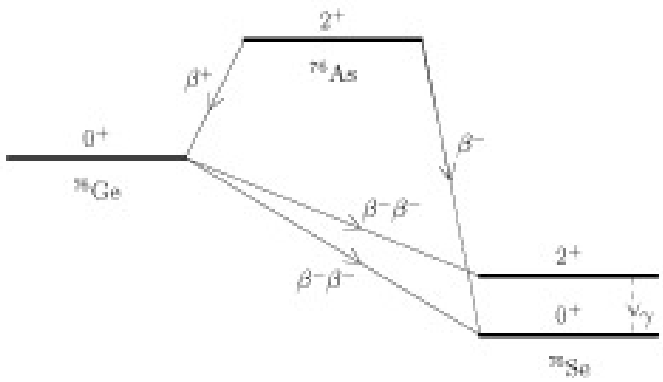

Figure 7.4: Schematic illustration of the energy level structure of the 2β-decay of ^{76}Ge into ^{76}Se [12].

Conversely, the neutrinoless double-beta decay ($0\nu\beta\beta$) processes are of the type:

$$\mathcal{N}(A, Z) \rightarrow \mathcal{N}(A, Z + 2) + 2e^- \tag{7.9}$$

Because no antineutrinos are emitted, the conservation of total lepton number is violated by two units ($\Delta L = 2$). This process is strictly forbidden in the Standard Model.

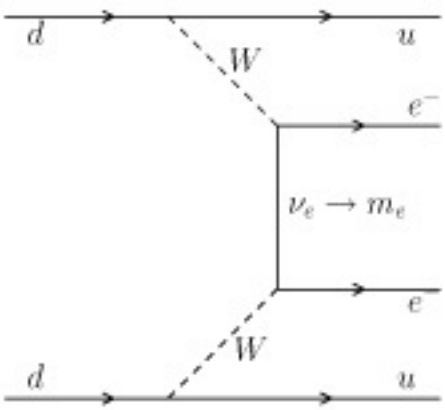

Figure 7.5: Tree-level diagram of the $0\nu\beta\beta$ process via the exchange of a light Majorana neutrino. The absence of outgoing neutrinos and the internal propagator line signify the violation of lepton number by two units ($\Delta L = 2$). Adapted from Giunti & Kim [12].

The Effective Majorana Mass and the Neutrino Propagator

In the Standard Model, the $0\nu\beta\beta$ decay is impossible due to two fundamental mismatch conditions at the weak interaction vertices:

1. **Particle-antiparticle mismatch:** An antineutrino ($\bar{\nu}_e$) emitted at the first leptonic vertex cannot be absorbed at the second vertex, which is only capable of absorbing a neutrino (ν_e).

2. **Helicity mismatch:** The weak interaction emits a neutral lepton with positive helicity, but the absorbing vertex requires a neutral lepton with negative helicity.

To satisfy these conditions, the electron neutrino must be a massive Majorana particle ($\bar{\nu}_e = \nu_e$ and $m_{\nu_e} \neq 0$). The Majorana nature resolves the particle-antiparticle mismatch, while the non-zero mass allows the neutrino to be emitted with the "wrong" negative helicity with a relative amplitude proportional to m_{ν_e}/E_{ν_e}.

This kinematic penalty is known as **helicity suppression**. Because the Standard Model weak interaction (V−A) strictly couples to left-handed particles and right-handed antiparticles, the neutrino is forced to undergo a helicity "flip" during the propagation between the two nucleons. The amplitude for the neutrino to be found in this "wrong" helicity state is strictly proportional to the ratio of its rest mass to its dynamic energy (m_ν/E_ν). Since the typical energy scale of the nuclear decay is $E_\nu \sim 35$ MeV, and the neutrino mass is $m_\nu \lesssim 0.1$ eV, this suppression factor is extraordinarily strong ($< 10^{-8}$). This severe helicity suppression is the primary physical reason why $0\nu\beta\beta$ is so exceptionally rare, even if neutrinos are indeed Majorana particles.

Taking into account three-neutrino mixing, the massive Majorana neutrinos propagate between the two leptonic weak interaction vertices. The amplitude of the $0\nu\beta\beta$ decay is governed by the effective Majorana mass:

$$m_{2\beta} = \sum_{k=1}^{3} U_{ek}^2 m_k \tag{7.10}$$

The explicit dependence on $m_{2\beta}$ is rigorously derived by evaluating the propagator describing the internal neutrino line. Using the properties of the charge conjugation

matrix $\mathcal{C}$ and the chiral projectors, the propagator is given by:

$$\begin{aligned} G(x_1 - x_2) &= \langle 0|T[\nu_{eL}(x_1)\nu_{eL}^T(x_2)]|0\rangle \\ &= \frac{1-\gamma^5}{2}\sum_{k=1}^{3} U_{ek}^2 \langle 0|T[\nu_k(x_1)\nu_k^T(x_2)]|0\rangle \left(\frac{1-\gamma^5}{2}\right)^T \\ &= -\frac{1-\gamma^5}{2}\sum_{k=1}^{3} U_{ek}^2 \langle 0|T[\nu_k(x_1)\overline{\nu_k}(x_2)]|0\rangle \mathcal{C}\frac{1-(\gamma^5)^T}{2} \\ &= -i\sum_{k=1}^{3} U_{ek}^2 \int \frac{d^4p}{(2\pi)^4}\frac{m_k}{p^2-m_k^2}e^{-ip\cdot(x_1-x_2)}\frac{1-\gamma^5}{2}\mathcal{C}. \end{aligned} \tag{7.11}$$

The average neutrino momentum in the nucleus is $\langle p^0\rangle \sim \langle|\vec{p}|\rangle \sim 35$ MeV, which is much larger than the light neutrino masses. Hence, the neutrino propagator can be approximated by factoring out the effective mass:

$$G(x_1 - x_2) \simeq -im_{2\beta}\int \frac{d^4p}{(2\pi)^4}\frac{e^{-ip\cdot(x_1-x_2)}}{p^2}\frac{1-\gamma^5}{2}\mathcal{C}. \tag{7.12}$$

Because the elements of the mixing matrix U_{ek} are complex, the terms in $m_{2\beta}$ can undergo partial or total cancellation. Using the standard parameterization of the PMNS matrix, the effective Majorana mass expands to:

$$m_{2\beta} = |U_{e1}|^2 m_1 + e^{i\alpha_2}|U_{e2}|^2 m_2 + e^{i\alpha_3}|U_{e3}|^2 m_3 \tag{7.13}$$

where α_2 and α_3 are the physically relevant Majorana CP-violating phases.

The Black Box Theorem

One might ask if the massive Majorana character of neutrinos is not only a sufficient, but also a necessary condition for $0\nu\beta\beta$-decay. The answer is unequivocally positive, as demonstrated by the Schechter-Valle "black box" theorem [12].

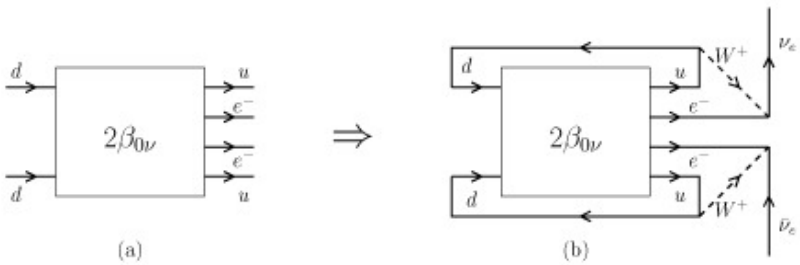

Figure 7.6: (a) Black-box diagram of the elementary $0\nu\beta\beta$ process $2d \to 2u + 2e$. (b) $\bar{\nu}_e \to \nu_e$ transition diagram routed through a $0\nu\beta\beta$ black-box [12].

If $0\nu\beta\beta$-decay occurs through any mechanism at any order in perturbation theory, we can represent it as a generic black box (Figure 7.6a). By joining the external electron and quark lines with standard W boson vertices, we obtain a diagram for a $\bar{\nu}_e \to \nu_e$ transition (Figure 7.6b). This 4-loop diagram necessarily generates a Majorana mass term for the electron neutrino through radiative corrections. Because no symmetry can forbid this generated mass without also forbidding the initial $0\nu\beta\beta$ decay, the observation of $0\nu\beta\beta$ rigorously guarantees that neutrinos are massive Majorana particles.

Phenomenology and Nuclear Matrix Elements

The $0\nu\beta\beta$ half-life of a nucleus $\mathcal{N}$ is given by:

$$[T_{1/2}^{0\nu}(\mathcal{N})]^{-1} = G_{0\nu}^{\mathcal{N}} |\mathcal{M}_{0\nu}^{\mathcal{N}}|^2 m_{2\beta}^2 \tag{7.14}$$

where $G_{0\nu}^{\mathcal{N}}$ is the phase space factor and $\mathcal{M}_{0\nu}^{\mathcal{N}}$ is the nuclear matrix element.

While the phase space factor can be calculated with high precision, the calculation of the nuclear matrix element $\mathcal{M}_{0\nu}^{\mathcal{N}}$ introduces the largest theoretical uncertainty in the field. Extracting $m_{2\beta}$ from a half-life limit requires complex many-body nuclear models (such as the Quasiparticle Random Phase Approximation, QRPA, or the Nuclear Shell Model).

For example, realistic calculations of $|\mathcal{M}_{0\nu}^{^{76}\mathrm{Ge}}|$ for Germanium-76 vary systematically between 1.50 and 4.59 depending on the specific model applied [12]. Consequently, when a modern experiment establishes a stringent lower bound on the half-life (e.g., $T_{1/2}^{0\nu} > 1.9 \times 10^{26}$ years), the resulting upper limit on the effective Majorana mass $|m_{2\beta}|$ is not a single number, but a broad range (e.g., $75 - 200$ meV) reflecting this exact theoretical uncertainty. Reducing the systematic errors in nuclear matrix element calculations is a critical priority for the future of neutrino physics.

7.5 Experimental Searches for Majorana Neutrinos

Because $0\nu\beta\beta$ is the singular practical test for the Majorana mass hypothesis, a massive global experimental effort is underway. The experimental signature is a monoenergetic peak at the exact Q-value of the decay, as the two emitted electrons carry away the entirety of the reaction's kinetic energy without any missing momentum from antineutrinos.

The LEGEND-200 experiment, utilizing 142.5 kg of solid germanium crystals enriched in ^{76}Ge, recently reported results based on 61.0 kg·yr of exposure [15]. Observing no signal above the background, they established a lower limit on the half-life of:

$$T_{1/2}^{0\nu} > 1.9 \times 10^{26} \text{ years} \quad (90\% \text{ C.L.}). \tag{7.15}$$

Depending on the nuclear matrix element calculations applied, this translates to an upper limit on the effective Majorana mass of:

$$m_{2\beta} < (75 - 200) \text{ meV}. \tag{7.16}$$

Complementary limits are continually provided by other leading experiments utilizing different isotopes to bypass specific nuclear matrix element uncertainties. The KamLAND-Zen experiment suspends xenon gas enriched in ^{136}Xe within a liquid scintillator to achieve massive target volumes, currently setting an even more stringent limit of:

$$T_{1/2}^{0\nu} > 2.3 \times 10^{26} \text{ years} \quad \Rightarrow \quad m_{2\beta} < (36 - 156) \text{ meV}. \tag{7.17}$$

Meanwhile, the CUORE experiment utilizes cryogenic tellurium dioxide (^{130}Te) bolometers operating at temperatures near absolute zero to search for the decay with extreme energy resolution [16].

7.6 Summary

In summary, the Dirac neutrino scenario conserves lepton number and requires a distinct antiparticle, while the Majorana scenario elegantly violates lepton number and treats the neutrino as its own antiparticle. These cases are experimentally distinguished entirely by the existence of $0\nu\beta\beta$ decay. Current generation experiments are setting incredibly stringent limits on the effective mass $m_{2\beta}$, and future ton-scale searches aim to fully cover the inverted hierarchy parameter space. If those future experiments observe a null result, it will place immense theoretical pressure on the Majorana hypothesis, leaving the Dirac mass structure as the favored explanation for the origin of neutrino mass.

Chapter 8

The Weinberg Dimension-5 Operator

8.1 The Effective Field Theory Framework and the Weinberg Operator

The Standard Model (SM) is highly successful at the electroweak scale, but it is fundamentally understood to be an effective field theory (EFT) valid only below some high-energy new-physics scale Λ. To capture the physics of this heavier scale at accessible energies, the full Lagrangian is expanded into a tower of non-renormalizable operators:

$$\mathcal{L}_{\text{EFT}} = \mathcal{L}_{\text{SM}} + \frac{1}{\Lambda}\sum_k c_k^{(5)}\mathcal{O}_k^{(5)} + \frac{1}{\Lambda^2}\sum_k c_k^{(6)}\mathcal{O}_k^{(6)} + \cdots , \tag{8.1}$$

where $\mathcal{O}_k^{(d)}$ are gauge- and Lorentz-invariant operators constructed from SM fields with mass dimension $d > 4$, and $c_k^{(d)}$ are dimensionless Wilson coefficients.

This approach of using non-renormalizable operators to predict high-scale physics has a famous historical precedent. In 1934, Enrico Fermi modeled nuclear beta decay as a dimension-6 four-fermion contact interaction, $\mathcal{L}_{\text{Fermi}} \propto G_F(\bar{p}\gamma^\mu n)(\bar{e}\gamma_\mu \nu_e)$. At low energies, this effective theory worked perfectly. However, the dimensional suppression of the coupling constant ($G_F \sim [\text{Mass}]^{-2}$) correctly predicted that the contact interaction was actually the low-energy limit of a heavy mediator exchange: the W boson ($G_F \sim g^2/M_W^2$). Today, we apply the exact same logic to neutrino masses, but starting with a dimension-5 operator to predict the heavy Seesaw mediators.

In 1979, Steven Weinberg mathematically demonstrated that, subject exclusively to the $SU(3)_c \times SU(2)_L \times U(1)_Y$ gauge symmetry and the known particle content of the SM, there is precisely *one* independent dimension-five operator [17]. This unique operator couples two left-handed lepton doublets (L) to two Higgs doublets (ϕ):

$$\mathcal{O}_5 = \frac{c_{\alpha\beta}}{\Lambda}\left(\overline{L_\alpha}\tilde{\phi}\right)\left(\tilde{\phi}^T L_\beta^c\right) + \text{h.c.}, \tag{8.2}$$

where $L_\alpha = (\nu_\alpha, \ell_\alpha)^T$ is the lepton doublet of flavor α, $\tilde{\phi} = i\tau_2\phi^*$ is the conjugate Higgs doublet, and $L^c = C\overline{L}^T$ denotes charge conjugation. Because the operator involves the contraction of identical fermionic fields, the dimensionless coefficient matrix $c_{\alpha\beta}$ is strictly symmetric by Fermi statistics [18].

Furthermore, if we assign a global lepton number $L = 1$ to the SM lepton doublets, the Weinberg operator transforms as $\mathcal{O}_5 \to e^{2i\alpha}\mathcal{O}_5$ under $U(1)_L$ phase rotations. It thus violates total lepton number by two units ($\Delta L = 2$), acting as the foundational mathematical trigger for Majorana mass generation [17, 18].

8.2 Mass Generation, Naturalness, and the High-Energy Scale

Following electroweak symmetry breaking (EWSB), the Higgs field acquires its vacuum expectation value $\langle\phi\rangle = (0, v/\sqrt{2})^T$. Substituting this fixed background field into Equation (8.2) yields a bare Majorana mass term for the left-handed neutrinos:

$$\boxed{(m_\nu)_{\alpha\beta} = \frac{c_{\alpha\beta}\, v^2}{\Lambda}.} \tag{8.3}$$

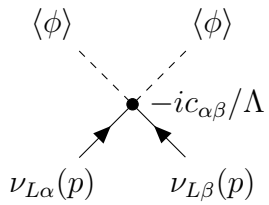

Figure 8.1: Effective Feynman diagram for the Weinberg dimension-5 operator after EWSB. The filled circle denotes the effective vertex $-ic_{\alpha\beta}/\Lambda$. The interaction with the dual Higgs vacuum expectation values generates the Majorana mass insertion $(m_\nu)_{\alpha\beta} = c_{\alpha\beta}v^2/\Lambda$ [17].

The parametric suppression of the neutrino mass by the ratio v^2/Λ is technically natural in the 't Hooft sense: setting $m_\nu \to 0$ artificially restores the global $U(1)_L$ symmetry of the Lagrangian.

Assuming the Wilson coefficients are of order unity ($c_{\alpha\beta} \sim \mathcal{O}(1)$), the observed absolute neutrino mass scale of $m_\nu \sim 0.1$ eV allows us to directly probe the scale of new physics:

$$\Lambda \sim \frac{v^2}{m_\nu} \sim \frac{(174\ \text{GeV})^2}{0.1\ \text{eV}} \sim 3 \times 10^{14}\ \text{GeV} \tag{8.4}$$

This remarkable result indicates that the mechanism responsible for neutrino mass generation naturally points to an energy scale strikingly close to the Grand Unified Theory (GUT) scale [19].

Crucially, the exact high-energy UV completion, whether it be the exchange of heavy right-handed neutrinos (Type-I Seesaw), heavy scalar triplets (Type-II), or fermion triplets (Type-III), is cleanly abstracted away by the EFT framework. However, the Schechter-Valle "black box" theorem guarantees that any definitive experimental observation of neutrinoless double-beta decay ($0\nu\beta\beta$) strictly implies a non-zero $(m_\nu)_{\alpha\beta}$ [20]. Such a discovery would unequivocally confirm the Majorana nature of neutrinos and prove that the Weinberg $\mathcal{O}_5$ operator is actively realized in nature, regardless of the specific high-scale mechanism that generates it.

Chapter 9

Seesaw Mechanisms

9.1 Overview of the Mechanism

The irrefutable experimental evidence for neutrino oscillations demands that active neutrinos possess non-zero mass, a feature strictly forbidden by the renormalizable Lagrangian of the Standard Model (SM). The most theoretically elegant framework to resolve this tension is the *seesaw mechanism*. By introducing heavy new-physics states and subsequently integrating them out below their mass threshold, this mechanism naturally generates the unique dimension-five Weinberg operator discussed in Section 8.

There exist exactly three renormalizable, tree-level ultraviolet (UV) completions of $\mathcal{O}_5$, differing fundamentally in the spin and the $SU(2)_L \times U(1)_Y$ gauge representations of the heavy mediators. While they universally yield a characteristic mass suppression driven by the heavy lepton-number-violating scale (M_{heavy}), the exact parameter dependence bifurcates based on the spin of the mediator:

For the **fermionic** mediators (Type I and Type III), the Weinberg operator is generated purely through Dirac Yukawa couplings (Y), resulting in a mass suppression strictly quadratic in Y:

$$m_\nu \sim \frac{Y^2 v^2}{M_{\text{heavy}}}. \tag{9.1}$$

Conversely, for the **scalar** mediator (Type II), the diagram requires both a Yukawa coupling (Y_Δ) to the leptons and a lepton-number-violating trilinear mass parameter (μ) coupling to the Higgs. This yields a light neutrino mass that is strictly linear in the Yukawa coupling:

$$m_\nu \sim \frac{\mu\, Y_\Delta v^2}{M_{\text{heavy}}^2}. \tag{9.2}$$

In all cases, v represents the electroweak vacuum expectation value. The coexistence of these distinct UV pathways highlights the versatility of the effective field theory framework.

9.2 Unpacking the Operator: The $SU(2)_L$ Algebra

To understand why there are exactly three tree-level completions, we must mathematically "unpack" the 4-point Weinberg operator ($LL\phi\phi$) into two 3-point vertices connected by a heavy mediator. Topologically, there are only two ways to pair these fields: coupling a lepton to a Higgs ($L \otimes \phi$) or coupling a lepton to a lepton ($L \otimes L$).

Path 1: The Fermionic Path ($L \otimes \phi$). Pairing a lepton doublet with a Higgs doublet creates a heavy intermediate fermion. Under $SU(2)_L$, both L and ϕ are doublets (denoted as $\mathbf{2}$). Their tensor product decomposes as:

$$\mathbf{2} \otimes \mathbf{2} = \mathbf{1} \oplus \mathbf{3} \tag{9.3}$$

This group theory constraint dictates that the heavy intermediate fermion must either be an $SU(2)_L$ singlet ($\mathbf{1}$) or a triplet ($\mathbf{3}$). The singlet case corresponds to the Type I seesaw (heavy right-handed neutrinos), while the triplet case corresponds to the Type III seesaw (heavy fermion triplets).

Path 2: The Scalar Path ($L \otimes L$). Pairing two lepton doublets creates a heavy intermediate scalar boson that eventually decays into two Higgs fields. The tensor product of two identical fermion doublets decomposes into antisymmetric and symmetric parts:

$$\mathbf{2} \otimes \mathbf{2} = \mathbf{1}_A \oplus \mathbf{3}_S \tag{9.4}$$

Because the resulting Majorana mass matrix must be strictly symmetric due to Fermi statistics, the antisymmetric singlet ($\mathbf{1}_A$) is forbidden. Therefore, the heavy scalar mediator is forced to be a symmetric triplet ($\mathbf{3}_S$). This corresponds to the Type II seesaw.

Thus, the $SU(2)_L$ algebra rigorously restricts the tree-level UV completions of neutrino mass to exactly three unique mediators.

9.3 Type I Seesaw: Heavy Right-Handed Neutrino Singlets

New Fields and Lagrangian

The Type I seesaw [21, 13, 14] is arguably the simplest and most heavily studied extension of the SM for neutrino mass generation. It augments the SM field content with n right-handed neutrino fields N_{Ri}, which are total singlets under the SM gauge group, transforming as $(\mathbf{1}, \mathbf{1}, 0)$. Because the observed solar and atmospheric mass splittings demand at least two massive active neutrinos, one strictly requires $n \geq 2$ singlets to reproduce the phenomenological data [19].

The introduction of these heavy singlets is strongly motivated by high-energy frameworks such as Grand Unified Theories (GUTs) which naturally predict new physics near the 10^{14} GeV scale derived in Section 8. The extended Lagrangian reads:

$$\mathcal{L}_{\mathrm{I}} = \mathcal{L}_{\mathrm{SM}} + i\overline{N_R}\partial\!\!\!/ N_R - \left[(Y_\nu)_{\alpha i}\, \overline{L_\alpha}\tilde{\phi}\, N_{Ri} + \frac{1}{2}\overline{N^c_{Ri}}\,(M_R)_{ij}\, N_{Rj} + \text{h.c.}\right], \tag{9.5}$$

where $(Y_\nu)_{\alpha i}$ is a $3 \times n$ complex Dirac Yukawa matrix coupling the SM lepton doublets L_α to N_{Ri} via the conjugate Higgs doublet $\tilde{\phi}$, and $(M_R)_{ij}$ is a symmetric $n \times n$ Majorana mass matrix. Crucially, because the N_{Ri} fields carry zero weak isospin and zero hypercharge, the bare Majorana mass term M_R is explicitly permitted by gauge invariance without requiring spontaneous symmetry breaking [**?**]. The structural co-existence of a symmetry-breaking Dirac mass and a gauge-invariant Majorana mass is the core engine of the seesaw.

The Mass Matrix and Seesaw Formula

After electroweak symmetry breaking (EWSB), the Higgs acquires its vacuum expectation value $\langle\phi\rangle = (0, v/\sqrt{2})^T$, yielding the Dirac mass matrix $m_D = Y_\nu v/\sqrt{2}$. If one imposes

strict lepton number conservation ($M_R = 0$), the matrix yields pure Dirac neutrinos. However, achieving sub-eV active neutrino masses would require unnaturally small, finely-tuned Yukawa couplings ($Y_\nu \sim 10^{-12}$). This makes the Majorana seesaw limit ($M_R \gg m_D$) aesthetically and theoretically preferred.

To find the physical mass eigenstates, one must analyze the Lagrangian in the basis of chiral states $\Psi_L = (\nu_L,\, N_R^c)^T$. In this basis, the complete $(3+n) \times (3+n)$ neutrino mass matrix takes the symmetric block form [**?**]:

$$\mathcal{M} = \begin{pmatrix} 0 & m_D \\ m_D^T & M_R \end{pmatrix}. \tag{9.6}$$

In the seesaw limit where the eigenvalues of M_R are vastly larger than the electroweak scale ($M_R \gg m_D$), block-diagonalization using a unitary transformation separates the physical spectrum into two distinct sectors [**?**]: 1. A heavy sterile sector composed predominantly of the right-handed singlets, with mass matrix roughly equal to M_R. 2. A light active sector composed predominantly of the $SU(2)_L$ flavor states, yielding the celebrated seesaw mass formula:

$$\boxed{m_\nu \simeq -m_D\, M_R^{-1}\, m_D^T = -\frac{v^2}{2}\, Y_\nu\, M_R^{-1}\, Y_\nu^T.} \tag{9.7}$$

The tree-level realization of the effective Weinberg operator via the intermediate N_R propagator is depicted in Figure 9.1.

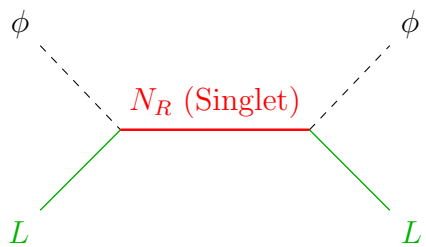

Figure 9.1: Tree-level Feynman diagram for the Type I seesaw mechanism. The dimension-five Weinberg operator $\mathcal{O}_5$ is mediated by the exchange of a heavy fermion singlet (the right-handed neutrino N_R). Topology and color scheme adapted from Strumia & Vissani [18].

Leptogenesis

A profound further virtue of the Type I seesaw is that it naturally accommodates a mechanism for generating the observed baryon asymmetry of the universe through *leptogenesis* [22].

In the early universe, at temperatures $T \sim M_R$, the heavy right-handed neutrinos are thermally produced and subsequently decay out of equilibrium into leptons and Higgs bosons ($N \to L\phi$ and $N \to \overline{L}\phi^\dagger$). Because these decays violate total lepton number and can violate CP, they generate a net lepton asymmetry, which non-perturbative electroweak sphaleron processes then partially convert into the observed baryon asymmetry.

The CP asymmetry ε_i generated by the decay of the i-th heavy state arises from the quantum interference between the tree-level decay amplitude and the one-loop (vertex and self-energy) corrections [18]. Assuming a hierarchical mass spectrum for the heavy

right-handed neutrinos ($M_1 \ll M_{2,3}$), the CP asymmetry parameter for the decay of the lightest state N_1 is given parametrically by:

$$\varepsilon_1 \simeq \frac{1}{8\pi} \sum_{j \neq 1} \frac{\mathrm{Im}\left[(Y_\nu^\dagger Y_\nu)_{1j}^2\right]}{(Y_\nu^\dagger Y_\nu)_{11}} \left(\frac{M_1}{M_j}\right). \tag{9.8}$$

This beautifully intertwines the high-energy CP-violating phases hidden within the neutrino Dirac Yukawa matrix (Y_ν) directly with the cosmic origins of matter.

Collider Phenomenology

In stark contrast to the triplet-based mechanisms, the high-energy collider phenomenology of the minimal Type I seesaw is notoriously pessimistic. Because the right-handed neutrinos N_R are complete Standard Model gauge singlets, they do not couple to the photon, the W boson, or the Z boson. Consequently, they cannot be directly produced via standard Drell-Yan processes at hadron colliders like the LHC.

Furthermore, if the Majorana mass scale M_R is near the GUT scale (10^{14} GeV) as motivated by naturalness, direct kinematic production at terrestrial colliders is fundamentally impossible. The heavy right-handed singlets are effectively "ghosts" at high energies, leaving the dimension-five Weinberg operator, and by extension, neutrinoless double-beta decay ($0\nu\beta\beta$), as the sole experimental probe of this minimal mechanism [19].

9.4 Type II Seesaw: Scalar $SU(2)_L$ Triplet

New Fields and Lagrangian

The Type II seesaw [23, 24] generates Majorana neutrino masses without the need for right-handed fermions. Instead, it expands the scalar sector by introducing a heavy complex scalar triplet Δ. To form a gauge-invariant Yukawa coupling with two left-handed SM lepton doublets (each possessing weak hypercharge $Y = -1$), the triplet must strictly transform under the SM gauge group as $(\mathbf{1}, \mathbf{3}, +2)$ [18].

This triplet consists of three physical complex scalar components: a neutral state Δ^0, a singly-charged state Δ^+, and a uniquely distinctive doubly-charged state Δ^{++}. The phenomenologically relevant additions to the Lagrangian are:

$$\mathcal{L}_{\mathrm{II}} = \mathcal{L}_{\mathrm{SM}} + \mathrm{Tr}[(D_\mu \Delta)^\dagger (D^\mu \Delta)] - M_\Delta^2 \,\mathrm{Tr}[\Delta^\dagger \Delta] - \Big[(Y_\Delta)_{\alpha\beta}\, \overline{L_\alpha^c}\, i\tau_2 \Delta\, L_\beta + \mu\, \phi^T i\tau_2 \Delta^\dagger \phi + \mathrm{h.c.}\Big]. \tag{9.9}$$

Here, the triplet Δ is represented as a 2×2 matrix, necessitating the trace operator (Tr) for the kinetic and mass terms. The matrix $(Y_\Delta)_{\alpha\beta}$ is a symmetric 3×3 Yukawa coupling matrix. The parameter μ is a mass-dimension trilinear coupling that connects the heavy scalar triplet directly to two SM Higgs doublets. Crucially, because the scalar triplet is assigned a lepton number of $L = -2$ to conserve L in the Yukawa interaction, the trilinear μ term explicitly (or softly) breaks lepton number conservation by two units, fulfilling the fundamental requirement for Majorana mass generation [18].

The Induced VEV and Seesaw Formula

The mechanism of mass generation in Type II relies on the interplay within the scalar potential. Upon electroweak symmetry breaking (EWSB), the SM Higgs doublet acquires

its standard vacuum expectation value $\langle\phi\rangle = (0, v/\sqrt{2})^T$. When this background field is substituted into the trilinear interaction term of Equation (9.9), it acts as a linear "tadpole" source for the neutral component of the triplet:

$$V(\phi, \Delta) \supset M_\Delta^2 |\Delta^0|^2 - \frac{\mu\, v^2}{2}(\Delta^0 + \Delta^{0*}). \tag{9.10}$$

Minimizing this effective potential with respect to Δ^0 reveals that the triplet is forced to acquire a small, induced vacuum expectation value:

$$v_\Delta \equiv \langle\Delta^0\rangle = \frac{\mu\, v^2}{2\, M_\Delta^2}. \tag{9.11}$$

This induced vev (v_Δ) is doubly suppressed relative to the electroweak scale by the ratio v^2/M_Δ^2. This parametric suppression naturally explains why v_Δ can be small enough (typically $\lesssim \mathcal{O}(1\text{ GeV})$) to evade precision electroweak constraints, such as the ρ parameter, while still successfully driving neutrino masses [19].

This small induced vev directly sources the Majorana neutrino masses through the triplet Yukawa coupling Y_Δ. By substituting $\langle\Delta^0\rangle = v_\Delta$ into the Yukawa term of the Lagrangian, one obtains the light active neutrino mass matrix:

$$\boxed{(m_\nu)_{\alpha\beta} = 2\,(Y_\Delta)_{\alpha\beta}\, v_\Delta = \frac{\mu\, v^2}{M_\Delta^2}\,(Y_\Delta)_{\alpha\beta}.} \tag{9.12}$$

Unlike the Type I and Type III formulas, which rely on the quadratic inversion of the Dirac mass matrix, the light neutrino mass in the Type II seesaw is strictly *linear* with respect to the associated Yukawa coupling Y_Δ. The tree-level realization of this mechanism is depicted in Figure 9.2.

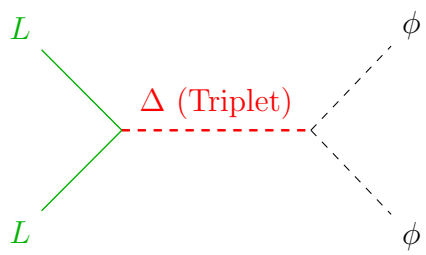

Figure 9.2: Tree-level Feynman diagram for the Type II seesaw mechanism. The dimension-five Weinberg operator $\mathcal{O}_5$ is mediated by the exchange of a heavy scalar $SU(2)_L$ triplet Δ. Topology and color scheme adapted from Strumia & Vissani [18].

Collider Phenomenology

Because the Type II scalar triplet carries both weak isospin and hypercharge, it strongly couples to the SM electroweak gauge bosons. Consequently, the heavy triplet states can be directly produced at hadron colliders via Drell-Yan processes (e.g., $q\bar{q} \to \gamma^*/Z^* \to \Delta^{++}\Delta^{--}$). The doubly-charged scalar Δ^{++} is of particular experimental interest, as it decays almost exclusively into a pair of same-sign leptons ($\Delta^{++} \to \ell_\alpha^+ \ell_\beta^+$), providing a spectacularly clean, essentially background-free signature at the LHC [19, 25, 26].

9.5 Type III Seesaw: Fermionic $SU(2)_L$ Triplet

New Fields and Lagrangian

The Type III seesaw [27] completes the classification of tree-level UV completions of the Weinberg operator by introducing $n \geq 2$ fermionic $SU(2)_L$ triplets N_i. To form a gauge-invariant Yukawa coupling with the SM lepton doublets, these new fermions must transform under the SM gauge group as $(\mathbf{1}, \mathbf{3}, 0)$.

Each triplet consists of three physical components: a neutral state N^3 and two charged states $N^\pm = (N^1 \mp iN^2)/\sqrt{2}$. The extended Lagrangian perfectly parallels the previous seesaw mechanisms:

$$\mathcal{L}_{\text{III}} = \mathcal{L}_{\text{SM}} + i\,\text{Tr}[\overline{N}\not{D}N] - \frac{1}{2}\,\text{Tr}\left[\overline{N^c}\,M_N\,N + \text{h.c.}\right] - \left[(\lambda^N)_{\alpha i}\,\overline{L_\alpha}\,\tau^a\tilde{\phi}\,N_i^a + \text{h.c.}\right]. \quad (9.13)$$

Here, M_N is the symmetric Majorana mass matrix for the heavy triplets, and λ^N is the Dirac Yukawa coupling. Because the N fields carry weak isospin, their kinetic term must utilize the gauge-covariant derivative ($\not{D}$). Furthermore, because the triplet carries zero hypercharge, the neutral component is its own antiparticle, naturally requiring the factor of 1/2 in the Majorana mass term to avoid double-counting degrees of freedom [18].

The Mass Matrix and Seesaw Formula

After electroweak symmetry breaking (EWSB), the Higgs acquires its vacuum expectation value $\langle\phi\rangle = (0, v/\sqrt{2})^T$. The neutral component N^3 then acquires a Dirac mass $m_D = \lambda^N v/\sqrt{2}$ connecting it to the SM active neutrinos.

Because N^3 is a Majorana fermion, the resulting mass matrix in the $(\nu_L, N^{3c})^T$ basis exhibits exactly the same block-symmetric structure as the Type I case (with the substitution $M_R \to M_N$):

$$\mathcal{M} = \begin{pmatrix} 0 & m_D \\ m_D^T & M_N \end{pmatrix}. \quad (9.14)$$

In the seesaw limit ($M_N \gg m_D$), block-diagonalization decouples the heavy triplet states and yields an identical light-neutrino mass formula:

$$\boxed{m_\nu = -m_D\,M_N^{-1}\,m_D^T = -\frac{v^2}{2}\,\lambda^N\,M_N^{-1}\,(\lambda^N)^T.} \quad (9.15)$$

The tree-level Feynman diagram illustrating this exchange is shown in Figure 9.3.

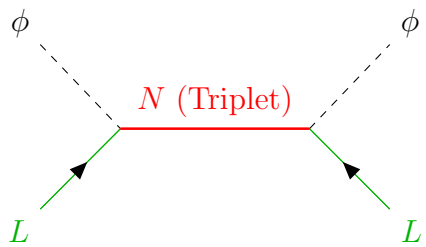

Figure 9.3: Tree-level Feynman diagram for the Type III seesaw mechanism. The dimension-five Weinberg operator $\mathcal{O}_5$ is mediated by the exchange of a heavy fermion $SU(2)_L$ triplet N. Topology and color scheme adapted from Strumia & Vissani [18].

Collider Phenomenology

While the light neutrino mass generation in Type III is mathematically degenerate with Type I, its high-energy phenomenology is drastically different. Because the physical $N^{\pm}$ and N^3 states couple directly to the photon and the W/Z bosons, they can be produced at hadron colliders like the LHC via s-channel Drell-Yan processes (e.g., $q\bar{q} \to W^{\pm} \to N^{\pm}N^3$). This provides a distinctive experimental signature that fundamentally distinguishes the triplet models from the pure singlet Type I scenario.

9.6 Comparative Summary

The three seesaw mechanisms detailed above constitute the strictly complete set of renormalizable, tree-level ultraviolet (UV) completions of the dimension-five Weinberg operator [19, 18]. While they differ fundamentally in the spin and gauge representations of their heavy mediators, they provide elegant, unified solutions to the origin of Majorana neutrino mass.

Mathematically, the fermionic completions (Types I and III) rely on the inversion of a heavy Majorana mass matrix, yielding a light active neutrino mass that scales quadratically with the Dirac Yukawa coupling. Conversely, the scalar completion (Type II) relies on an induced vacuum expectation value, yielding a strictly linear dependence on the Yukawa coupling.

Phenomenologically, the mechanisms drastically bifurcate at high energies. The gauge-singlet nature of the Type I mediator renders it essentially invisible to direct terrestrial collider searches. In stark contrast, the weak-triplet nature of the Type II and Type III mediators ensures direct coupling to the Standard Model electroweak gauge bosons, allowing for robust s-channel Drell-Yan production at the LHC. Finally, all three mechanisms intrinsically support high-scale leptogenesis, offering a profound theoretical framework capable of simultaneously explaining the extreme smallness of neutrino masses and the observed baryon asymmetry of the universe.

	Type I	**Type II**	**Type III**
New field	$N_R : (\mathbf{1}, \mathbf{1}, 0)_F$	$\Delta : (\mathbf{1}, \mathbf{3}, +2)_S$	$N : (\mathbf{1}, \mathbf{3}, 0)_F$
ΔL breaking	M_R (explicit)	μ (soft/explicit)	M_N (explicit)
Light mass formula	$-m_D M_R^{-1} m_D^T$	$\mu v^2 Y_\Delta / M_\Delta^2$	$-m_D M_N^{-1} m_D^T$
m_ν vs. Y	quadratic	linear	quadratic
EW gauge production	No	Yes (γ, W, Z)	Yes (γ, W, Z)
Leptogenesis	Yes	Yes	Yes
Key references	[21, 13, 14]	[23, 24]	[27]

Table 9.1: Comparison of the three tree-level seesaw mechanisms. Gauge representations are given under $SU(3)_c \times SU(2)_L \times U(1)_Y$; subscripts F/S denote fermionic or scalar mediators [19, 18].

Chapter 10

Sterile Neutrinos

10.1 Gauge Singlets and Helicity Suppression

A *sterile neutrino* (N_R) is defined as a fermion that is a complete singlet under the Standard Model (SM) gauge group, transforming as $(\mathbf{1}, \mathbf{1}, 0)$. Because it carries no color, weak isospin, or hypercharge, a sterile neutrino does not couple to the W, Z, or γ bosons. It can only interact with the Standard Model fields via mixing with the active neutrino sector.

As derived in Section 9 (Type I Seesaw), these sterile gauge singlets mix with active left-handed neutrinos via a Dirac mass term (m_D) to form a comprehensive mass matrix. However, because sterile neutrinos carry no weak charge, their direct detection via standard weak interactions requires a **helicity flip**. The quantum mechanical amplitude for a neutrino to undergo this helicity flip during an interaction is strictly proportional to the ratio of its mass to its energy (m_ν/E).

For standard terrestrial experiments observing MeV to GeV neutrinos, this introduces a severe kinematic suppression. The probability of the helicity flip is suppressed by a factor of roughly $\sim 10^{-12}$. Consequently, sterile states are nearly invisible in standard oscillation experiments unless their mixing with active states is anomalously large.

10.2 The Three Mass Regimes

The phenomenology of the extended active-sterile mass matrix $\mathcal{M}$ is defined by the relative sizes of the bare active Majorana mass (m), the Dirac mass (m_D), and the heavy sterile Majorana mass (M_R). This mathematical structure yields three distinct physical regimes:

- **Regime 1 (Decoupled):** $m_D \ll m, M_R$. In this limit, there is effectively no mixing between the active and sterile sectors. The sterile neutrinos completely decouple from the Standard Model, resulting in no observable oscillation effects.

- **Regime 2 (Pseudo-Dirac):** $m_D \gg m, M_R$. The active and sterile states form nearly degenerate pairs characterized by maximal mixing. To avoid conflicting with the stringent limits from current solar and atmospheric oscillation data, the mass scales in this regime are strictly constrained to $m, M_R \lesssim 10^{-9}$ eV.

- **Regime 3 (Seesaw):** $m \ll m_D \ll M_R$. This corresponds to the classic Type I seesaw limit (typically with $m = 0$). It naturally yields light, predominantly active neutrino masses of $\sim m_D^2/M_R$ and heavy, highly sterile states of mass $\sim M_R$.

10.3 eV-Scale Sterile Neutrinos and Short-Baseline Anomalies

Historically, experimental anomalies have motivated the search for light, eV-scale sterile neutrinos operating outside the standard three-flavor paradigm. The minimal extension to accommodate such a state is the $3+1$ scheme: three active neutrinos plus one sterile neutrino, introducing a new mass-squared splitting $\Delta m_{41}^2 \sim 1$ eV^2.

In this scheme, the short-baseline oscillation probability ($L/E \sim 1$ m/MeV) for ν_e appearance in a ν_μ beam is given by:

$$P(\nu_\mu \to \nu_e) \approx \sin^2 2\theta_{\mu e} \sin^2\left(\frac{\Delta m_{41}^2 L}{4E}\right), \quad \text{where} \quad \sin^2 2\theta_{\mu e} \equiv 4|U_{e4}|^2|U_{\mu 4}|^2. \tag{10.1}$$

Anomalous ν_e appearance signals observed at the LSND [28] and MiniBooNE [29] experiments were historically interpreted as evidence for such an eV-scale sterile state. However, the global experimental picture is highly inconsistent. Critically, independent high-precision searches from experiments like MicroBooNE [30] have found no evidence for the anomalous ν_e excess. Consequently, the eV-scale sterile neutrino hypothesis faces mounting experimental pressure and is widely considered to be in extreme tension with null results.

10.4 Cosmology and Dark Matter Constraints

Cosmology provides some of the most stringent bounds on the existence of sterile neutrinos. If light sterile neutrinos (mass $\lesssim 1$ eV) exist and possess sufficient mixing with active states, they would have been fully thermalized in the early universe. They would thus behave as extra relativistic radiation during the epochs of Big Bang Nucleosynthesis (BBN) and recombination.

This extra radiation is parameterized by its contribution to the effective number of relativistic species (N_{eff}):

$$N_{\text{eff}} = N_\nu^{\text{SM}} + \Delta N_{\text{eff}}, \quad \text{where} \quad N_\nu^{\text{SM}} = 3.046. \tag{10.2}$$

Observations of the Cosmic Microwave Background by the Planck satellite, combined with BBN abundance data, stringently constrain this value to $\Delta N_{\text{eff}} < 0.3$ at the 95% confidence level [31, 16]. This robustly rules out the existence of a fully thermalized, additional light sterile neutrino.

Sterile Neutrinos as Warm Dark Matter

Alternatively, a heavier sterile neutrino with a mass in the range of $m_s \sim 1 - 100$ keV serves as a highly motivated candidate for **Warm Dark Matter** (WDM). Such particles could be produced in the early universe via active-sterile mixing through the Dodelson-Widrow mechanism.

However, because these states are not entirely stable, they would slowly decay radiatively ($\nu_s \to \nu_a + \gamma$), producing a distinctive astrophysical signature: an X-ray emission line at exactly half the sterile neutrino mass ($E_\gamma = m_s/2$). Heavy constraints from X-ray observatories and analyses of the Lyman-α forest (which probes small-scale cosmic structure) have largely ruled out simple Dodelson-Widrow production, requiring more complex mechanisms (such as resonant production via lepton asymmetries) to remain viable dark matter candidates.

Ultimately, while the cosmological constraints on the sum of neutrino masses ($\sum m_\nu$) apply rigidly to the active sector, sterile neutrinos are additionally bounded by the limits on N_{eff} and their impact on cosmic structure formation.

Chapter 11

Cosmological Constraints on Neutrino Mass

11.1 The Cosmic Neutrino Background and Relic Density

To understand how massive neutrinos suppress the growth of cosmic structure, we must first determine their present-day energy density. In the early universe, the energy density of a relativistic species at temperature T is given by the standard phase-space integral [32]:

$$\rho(T) = \frac{g}{(2\pi)^3} \int \frac{4\pi p^3}{e^{p/T} \mp 1} dp = \begin{cases} \frac{\pi^2}{30} g T^4 & \text{(Bosons)} \\ \frac{7}{8}\frac{\pi^2}{30} g T^4 & \text{(Fermions)} \end{cases} \tag{11.1}$$

where g is the number of internal degrees of freedom. In the early universe, neutrinos were kept in thermal equilibrium with the primordial plasma via weak interactions, which have an interaction rate scaling as $\Gamma_W \sim G_F^2 T^5$. As the universe expanded, the weak interaction rate Γ_W dropped below the Hubble expansion rate H at a temperature of $T_{\text{dec}} \sim 1$ MeV. At this point, neutrinos decoupled to form the Cosmic Neutrino Background (CνB) [33].

Subsequent electron-positron annihilation heated the photon bath but not the decoupled neutrinos. The total comoving entropy $S \propto g_{*S} T^3 a^3$ is conserved during the expansion. Before e^+e^- annihilation ($T > m_e$), photons, electrons, and positrons are in equilibrium, giving the effective entropy degrees of freedom:

$$g_{*S}^{\text{before}} = g_\gamma + \frac{7}{8}(g_{e^-} + g_{e^+}) = 2 + \frac{7}{8}(2+2) = \frac{11}{2}. \tag{11.2}$$

After annihilation ($T < m_e$), only photons remain in the electromagnetic thermal bath, so $g_{*S}^{\text{after}} = g_\gamma = 2$. Because neutrinos had already decoupled, they did not share in this entropy transfer. Entropy conservation dictates the current temperature ratio between the Cosmic Microwave Background (CMB) and the CνB to be:

$$T_\nu = \left(\frac{g_{*S}^{\text{after}}}{g_{*S}^{\text{before}}}\right)^{1/3} T_\gamma = \left(\frac{4}{11}\right)^{1/3} T_\gamma \approx 1.95 \text{ K} \approx 1.7 \times 10^{-4} \text{ eV}. \tag{11.3}$$

Using Fermi-Dirac statistics for the neutrinos and Bose-Einstein for the CMB photons, the ratio of their number densities is exactly:

$$\frac{n_\nu}{n_\gamma} = \left(\frac{3}{4}\right)\left(\frac{T_\nu}{T_\gamma}\right)^3 = \left(\frac{3}{4}\right)\left(\frac{4}{11}\right) = \frac{3}{11}. \tag{11.4}$$

Given the measured CMB photon density $n_\gamma \approx 411$ cm^{-3}, the present-day number density per neutrino flavor is:

$$n_{\nu,i} = \frac{3}{4}\left(\frac{T_\nu}{T_\gamma}\right)^3 n_\gamma \approx 112 \text{ cm}^{-3}. \tag{11.5}$$

Because the current temperature of the universe is extremely low ($T_\nu \sim 1.7\times10^{-4}$ eV), at least two of the neutrino mass eigenstates are completely non-relativistic today. Their total energy density is given by their rest mass multiplied by their number density: $\rho_\nu = \sum_i m_i n_{\nu,i}$.

Dividing this by the critical density of the universe ($\rho_c = 3H_0^2/8\pi G$) yields the neutrino fractional contribution to the cosmic energy budget:

$$\Omega_\nu = \frac{\rho_\nu}{\rho_c} = \frac{\sum m_\nu}{93.14 \text{ eV} \cdot h^2}. \tag{11.6}$$

From this, we define the critical parameter f_ν, which is the neutrino fraction of the total matter density: $f_\nu \equiv \Omega_\nu/\Omega_m$ [34].

11.2 Free-Streaming and the Fluid Equations

Unlike cold dark matter (CDM), which decouples when it is highly non-relativistic and has essentially zero thermal velocity, neutrinos are hot/warm relics. They decouple while highly relativistic ($T \gg m_\nu$) and retain a large thermal velocity dispersion. As the universe expands, this velocity simply redshifts away inversely with the scale factor a:

$$v_{\text{th}} \propto \frac{1}{a}. \tag{11.7}$$

Because of this immense thermal velocity, neutrinos do not stay put. This high velocity allows neutrinos to "free-stream" out of small gravitational potential wells, physically acting as a cosmic "eraser" that completely washes out any neutrino density perturbations on small scales.

To quantify this boundary mathematically, we use the comoving free-streaming wavenumber k_{fs}. By definition, a "comoving" scale factors out the expansion of the universe, allowing us to track the same physical structure over cosmic time [**?**]. This defines the boundary below which neutrinos can no longer cluster [34]:

$$k_{\text{fs}}(a) = \sqrt{\frac{3}{2}}\frac{aH(a)}{v_{\text{th}}(a)}. \tag{11.8}$$

For small spatial scales (large wavenumbers where $k \gg k_{\text{fs}}$), neutrinos travel too fast to be captured. Their density contrast vanishes ($\delta_\nu \to 0$). To rigorously quantify the effect of this free-streaming on structure formation, we model CDM as a pressureless perfect fluid in the Newtonian gauge deep in the matter-dominated era. The sub-horizon evolution of the CDM density contrast, $\delta_c \equiv \delta\rho_c/\bar{\rho}_c$, is governed by the continuity and Euler equations [32]:

$$\dot{\delta}_c + \frac{1}{a}\vec{\nabla}\cdot\vec{v}_c = 0, \tag{11.9}$$

$$\dot{\vec{v}}_c + H\vec{v}_c = -\frac{1}{a}\vec{\nabla}\Phi. \tag{11.10}$$

Taking the time derivative of the continuity equation and substituting the divergence of the Euler equation yields the master evolution equation:

$$\ddot{\delta}_c + 2H\dot{\delta}_c = \frac{1}{a^2}\nabla^2\Phi. \tag{11.11}$$

The system is closed using the Poisson Equation for gravity, which connects the gravitational potential Φ to the *total* matter perturbation δ_m:

$$\frac{1}{a^2}\nabla^2\Phi = 4\pi G\bar{\rho}_m\delta_m. \tag{11.12}$$

11.3 Suppressed Growth of Cold Dark Matter

The total matter perturbation is the density-weighted sum of CDM and neutrinos: $\delta_m = (1 - f_\nu)\delta_c + f_\nu\delta_\nu$.

We now restrict our analysis to small scales ($k \gg k_{\text{fs}}$). In this regime, neutrino free-streaming completely washes out their density perturbations, meaning $\delta_\nu \to 0$. Therefore, the source of gravity in the Poisson equation is "starved" of the neutrino mass fraction, reducing to $\delta_m \approx (1 - f_\nu)\delta_c$. Substituting this into Equation (11.11) yields:

$$\ddot{\delta}_c + 2H\dot{\delta}_c = 4\pi G\bar{\rho}_m(1 - f_\nu)\delta_c. \tag{11.13}$$

During the matter-dominated era, the scale factor evolves as $a \propto t^{2/3}$. This allows us to write the Hubble parameter as $H = 2/(3t)$ and the background density as $4\pi G\bar{\rho}_m = \frac{3}{2}H^2 = 2/(3t^2)$. Substituting these relations yields a standard Euler-Cauchy differential equation [34]:

$$\ddot{\delta}_c + \frac{4}{3t}\dot{\delta}_c - \frac{2}{3t^2}(1 - f_\nu)\delta_c = 0. \tag{11.14}$$

We seek power-law solutions of the form $\delta_c \propto t^\alpha$. Substituting this ansatz into the differential equation gives the characteristic polynomial:

$$\alpha^2 + \frac{1}{3}\alpha - \frac{2}{3}(1 - f_\nu) = 0. \tag{11.15}$$

Solving for the growing mode α_+ using the quadratic formula:

$$\alpha_+ = \frac{-1/3 + \sqrt{1/9 + \frac{8}{3}(1 - f_\nu)}}{2} = \frac{-1 + \sqrt{25 - 24f_\nu}}{6}. \tag{11.16}$$

Because the neutrino mass fraction is small ($f_\nu \ll 1$), we apply a Taylor expansion to the square root ($\sqrt{1 - x} \approx 1 - x/2$):

$$\sqrt{25\left(1 - \frac{24}{25}f_\nu\right)} \approx 5\left(1 - \frac{12}{25}f_\nu\right) = 5 - \frac{12}{5}f_\nu. \tag{11.17}$$

Substituting this back into the solution for α_+:

$$\alpha_+ \approx \frac{-1 + 5 - \frac{12}{5}f_\nu}{6} = \frac{2}{3} - \frac{2}{5}f_\nu. \tag{11.18}$$

Since $t \propto a^{3/2}$, we convert this temporal growth rate back to the scale factor. The growth of small-scale CDM perturbations is analytically given by:

$$\delta_c(k \gg k_{\rm fs}) \propto a^{1-\frac{3}{5}f_\nu}. \tag{11.19}$$

The CDM perturbations grow slower than the standard $\delta_c \propto a$ because the universe is expanding at a rate dictated by the total matter density $\bar{\rho}_m$, but the gravitational wells driving the collapse are only sourced by the fraction $(1 - f_\nu)$.

11.4 Deriving the Power Spectrum Suppression

Cosmological surveys do not measure the CDM perturbation δ_c directly; they measure the total matter power spectrum, defined as $P_m(k) \propto |\delta_m(k)|^2$. The relative suppression of this spectrum compared to a pure ΛCDM model without massive neutrinos ($f_\nu = 0$) requires two distinct physical corrections.

First, we calculate the integrated growth suppression of CDM. From the time the relevant modes enter the horizon (a_i) to today ($a_0 = 1$), the relative suppression of the CDM power spectrum is:

$$\frac{P_c^{(f_\nu)}}{P_c^{(0)}} = \left(\frac{a_0}{a_i}\right)^{-\frac{6}{5}f_\nu} \approx 1 - \frac{6}{5}f_\nu \ln\left(\frac{a_0}{a_i}\right). \tag{11.20}$$

Second, we must correct for the baseline clustering penalty. Because neutrinos do not cluster on these small scales, the observable total matter perturbation is $\delta_m \approx (1 - f_\nu)\delta_c$. Squaring this yields the relationship between the observable matter spectrum and the CDM spectrum:

$$P_m(k) \propto (1 - f_\nu)^2|\delta_c|^2 \approx (1 - 2f_\nu)P_c(k). \tag{11.21}$$

Combining the direct clustering penalty ($-2f_\nu$) with the integrated fractional change in the CDM growth, the total observable matter power spectrum suppression is:

$$\frac{\Delta P_m(k)}{P_m(k)} \approx -2f_\nu - \frac{6}{5}f_\nu \ln\left(\frac{a_0}{a_i}\right). \tag{11.22}$$

For the length scales probed by actual linear Large Scale Structure surveys (e.g., $k \sim 0.1$ h/Mpc), evaluating the cosmic expansion history from horizon entry to today yields an effective logarithmic integral of roughly $\ln(a_0/a_i) \approx 5$ [34].

We can explicitly derive this scale factor a_i. A neutrino transitions from relativistic to non-relativistic when its average thermal momentum ($\langle p \rangle \approx 3.15T_\nu$) drops below its rest mass. Given the present-day neutrino temperature $T_{\nu,0} \approx 1.7 \times 10^{-4}$ eV, the temperature scales with redshift as $T_\nu(z) = T_{\nu,0}(1 + z)$. For a heaviest neutrino mass eigenstate of $m_3 \approx 0.05$ eV, equating momentum to mass yields a non-relativistic transition at $z \approx 100$, corresponding to a scale factor of $a_i \approx 0.01$. The number of "e-folds" of growth since that time is $\ln(1/0.01) \approx 4.6 \approx 5$, perfectly consistent with the suppression factor used in Large Scale Structure surveys.

Plugging this value into our equation yields the foundational analytical result:

$$\frac{\Delta P_m}{P_m} \approx -2f_\nu - \frac{6}{5}(5)f_\nu = -8f_\nu. \tag{11.23}$$

By substituting the definition of f_ν, we arrive at the absolute master formula used by modern cosmological surveys to constrain the neutrino mass:

$$\frac{\Delta P(k)}{P(k)} \approx -8\left(\frac{\sum m_\nu}{\Omega_m h^2 \cdot 93.14 \text{ eV}}\right). \tag{11.24}$$

11.5 Phenomenological Bounds and the Hierarchy Tension

Equation (11.24) establishes how $\sum m_\nu$ dictates the amplitude of structure suppression. To extract this mass from observational data, cosmologists measure the power spectrum using galaxy clustering and the weak gravitational lensing of the CMB. However, extracting $\sum m_\nu$ requires breaking degeneracies with other cosmological parameters, notably the late-time clustering amplitude (σ_8) and the Hubble constant (H_0).

Historically, Planck 2018 data combined with Baryon Acoustic Oscillations (BAO) constrained the sum of neutrino masses to $\sum m_\nu < 0.12$ eV (95% C.L.) [31]. However, recent generation-IV surveys have tightened this window dramatically. In 2024, the Dark Energy Spectroscopic Instrument (DESI) collaboration combined their highly precise Year 1 BAO measurements with Planck PR4 CMB data. By firmly locking down the expansion history, they broke the remaining geometric degeneracies and drove the upper bound down to [35]:

$$\sum m_\nu < 0.072 \text{ eV} \quad (95\% \text{ C.L.}). \tag{11.25}$$

This phenomenological result places extreme tension on terrestrial neutrino oscillation paradigms. Terrestrial measurements of Δm^2_{21} and $|\Delta m^2_{31}|$ dictate a minimum possible mass sum of $\sum m_\nu \approx 0.06$ eV for the Normal Ordering ($m_1 < m_2 \ll m_3$), and $\sum m_\nu \approx 0.10$ eV for the Inverted Ordering ($m_3 \ll m_1 < m_2$).

Consequently, the recent DESI+Planck bound of 0.072 eV robustly excludes the entire parameter space of the Inverted Mass Ordering. Unless there are unrecognized systematic errors in the data or extensions to the standard ΛCDM model, cosmology currently provides the strongest empirical evidence that nature has chosen the Normal Mass Hierarchy.

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